

COST AWARE STOPPING BOUNDARIES IN REASONING LANGUAGE MODELS

by

Aditya Bhatt

A thesis submitted to Johns Hopkins University
in conformity with the requirements for the degree of
Master of Science in Applied and Computational Mathematics

Baltimore, Maryland

October 2026

Abstract

Additional reasoning can repair an incorrect answer, replace a correct answer, or consume computation without sufficient improvement. This thesis formulates response-level stopping as a finite-horizon decision based on observable prefixes and hidden correctness. It proves the binary repair-corruption drift identity and the Bellman/Snell optimal stopping result, gives a sufficient persistence condition for a drift-sign rule, and constructs exact counterexamples to unconditional myopic optimality. The experimental record separates a variable-horizon matrix from a standardized corpus of 144,440 saved rows, 28,888 trajectories, and 2,948 tasks. Recomputed cluster-based tables show model- and domain-dependent continuation value and matched effects of estimator, token-cap, and precision changes. The historical stacked ROC-AUC of 0.955156 is retained as a retrospective non-nested diagnostic, rather than an online performance guarantee. A fitted prefix controller enforces a two-step floor and prevents future generation after stopping. On 100 new GSM8K questions, its actual arm saves 56.51 percent of completion tokens with 7 correct answers versus 6 at the full horizon. On 20 traps it saves 52.11 percent, with one correct answer in each arm. Every learned run stops at step two; adaptive benefit and accuracy noninferiority remain unestablished. The results support causal, cost-sensitive stopping experiments while leaving stronger calibration, universal optimality, and external noninferiority claims unestablished.

Research draft v1.0 for committee review. Defense, approval, final PDF/A validation, and institutional acceptance are pending.

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Chapter 1 Introduction and motivation

1.1 Reasoning as a sequential allocation problem

Language models can answer mathematical and scientific questions by producing intermediate text before committing to an answer. Chain-of-thought prompting makes that intermediate work explicit [1]. The resulting sequence can contain useful calculations, checks, and repairs. It can also contain repetition, an incorrect reinterpretation of the problem, or a revision that replaces a correct candidate with an incorrect one. Additional generation therefore has an uncertain benefit and a measurable cost. A stopping policy must decide when the expected benefit of another increment justifies that cost.

The problem is especially clear when a model supplies several successive candidate answers to the same question. A trajectory may begin incorrectly, recover, and then become incorrect again. Its final answer alone conceals both the earlier opportunity to stop and the risk of stopping before a later repair. Comparing only the first and last answers similarly conceals the sequence of changes. The appropriate observational unit is a trajectory with ordered candidate answers, and the relevant decision is made using the prefix available at the moment of stopping.

This thesis studies that decision in a deliberately controlled setting. Each reasoning increment is a complete model response containing a candidate answer. The model can be asked to revise that response at the next increment. The experiments are therefore about repeated response-and-revision trajectories. They should not be interpreted as observations of every latent thought, every token of an uninterrupted internal reasoning

process, or the model's unobserved cognition. This distinction determines what a controller can interrupt and how savings must be measured.

The motivating application is economical inference: spend computation when it improves the answer enough to be worthwhile, and terminate when further computation has low expected value. Economical inference is not identical to maximizing accuracy at any cost. A policy can improve a cost-sensitive utility while reducing accuracy; conversely, a policy can increase accuracy while increasing total computation. Both outcomes must be reported. Describing either one solely as a successful stop rate would hide the actual trade-off.

1.2 Operational meaning of overthinking

Let a saved candidate be correct when it satisfies the benchmark's frozen reference grader. A repair is an incorrect-to-correct transition between successive saved candidates. A corruption is a correct-to-incorrect transition. These terms describe an observable change in the answer label. They do not assert that the model was internally uncertain or that its narrative accurately reports its reasoning.

Three phenomena must be distinguished. First, **answer corruption** means that a later candidate is wrong after an earlier candidate was correct. Second, **unproductive continuation** means that additional generation fails to produce enough improvement to offset its cost. Third, **policy regret** means that a stopping decision has lower utility than a specified comparator. Corruption can occur without large regret when it is rare or inexpensive to repair. Unproductive continuation can occur even when accuracy is constant. Regret can arise because the policy stops before a repair, not only because it continues beyond a correct answer.

Published work on overthinking motivates the possibility that longer reasoning is inefficient [8]. This thesis asks a narrower, measurable question: on the recorded model-domain panels, when does another response increment have negative net value, and how much of that information can a causal controller use? The literature supplies context; the project's measured effects come from its own frozen artifacts.

At the population level, an accuracy curve can show where the average benefit of continuation changes sign. That curve is not a per-question stopping oracle. An early difficulty assessment, answer stability, or model confidence may differentiate questions that share the same average curve. Conversely, two indistinguishable prefixes may have different unobserved continuations. A scientifically useful controller must acknowledge both possibilities rather than treating the panel's best step as a universally correct stopping time.

1.3 Research questions

The first question is descriptive: do repair and corruption transitions explain the empirical change in correctness, after accounting for computation cost? The answer requires explicit denominators. Repair probability is computed among candidates currently wrong, and corruption probability among candidates currently correct. Dividing both counts by all candidates estimates transition frequencies, not their conditional probabilities. Chapter 2 derives the exact relationship between these quantities.

The second question is theoretical: when is a drift-sign rule an optimal stopping rule? A one-step improvement criterion is attractive because it is simple and cheap, but a negative immediate gain need not rule out a valuable later repair. This thesis gives the general finite-horizon solution, identifies a sufficient persistence condition under which a

myopic rule becomes optimal, and gives counterexamples when that condition fails. The theory is intended to make an engineering approximation understandable, not to attach an unconditional optimality claim to a convenient heuristic.

The third question is empirical: how do model family, scale, domain, temperature, token cap, and numerical precision relate to the observed trade-off? A cross-model association does not isolate parameter count, and a matched precision comparison does not prove that every model responds to quantization in the same way. Chapter 3 states the actual designs and Chapter 4 reports conclusions at their supported scope.

The fourth question is operational: can a controller make a prefix-only decision quickly enough to be used in a generation loop, and does that loop actually avoid future generation? These are separate checks. A function that returns a stop flag after all responses have already been generated is a replay evaluator. A live loop must cease requesting future increments, preserve the chosen answer, and record what was generated before termination. Chapter 5 distinguishes controller latency, live completion tokens, and retrospective counterfactual tokens.

1.4 Contributions

This thesis contributes an evidence-preserving experimental record that separates a variable-horizon boundary corpus from a standardized five-step detector corpus. The separation prevents double counting and keeps the model roster, benchmark split, sample size, and response horizon attached to the correct claim. A data manifest binds the selected artifacts to content hashes and records the software provenance that can actually be recovered.

Its mathematical contribution is a complete finite-horizon stopping formulation for observable reasoning prefixes with hidden correctness labels. The derivation separates conditional beliefs from panel accuracies, conditional transition probabilities from unconditional transition frequencies, and optimal continuation value from immediate drift. It also specifies what an error bound or a confidence claim would need to assume before it can justify a stopping decision.

Its empirical contribution is a set of reproducible descriptive and controlled analyses with uncertainty reported over the experimental cluster. The analyses include population repair and corruption curves, matched estimator comparisons, paired token-cap and precision contrasts, task-grouped detector evaluation, and explicit distinctions between ranking, calibration, and stopping utility. Negative results remain part of the argument because they constrain plausible explanations and deployment choices.

Its engineering contribution is an online stopping interface that admits only observations available at the current step, enforces the minimum response floor, and terminates generation through an explicit loop. Executable checks examine future-step independence, terminal behavior, token accounting, and adversarial trajectories. The engineering result establishes an implementable causal interface; statistical effectiveness is assessed through the separate measurements reported in Chapter 5.

1.5 Evidence and reproducibility

A hash verifies that a later artifact has the same selected content as the frozen artifact, subject to the manifest's stated byte-normalization rule. It does not verify that the original observations were generated honestly, that the labels are correct, that the selected sample is representative, or that the original experiment can be repeated on

arbitrary hardware. Those issues require provenance, software and hardware records, executable analyses, and independent evaluation.

This repository's Windows checkout can transform line endings. The freeze therefore distinguishes exact local bytes from canonical LF content and checks the canonical content against the historical tournament hashes. This is a reproducibility issue with a precise solution, not permission to ignore arbitrary changes. Both identities are retained, and verification must fail on an unexplained content difference.

Software provenance has a similar limitation. A lock file generated on the current workstation records the current environment. It cannot retrospectively establish the complete environment of the Blackwell tournament. The thesis reports historical versions where stored metadata supports them and identifies unavailable details. It avoids a claim that installing the workstation's lock will reproduce a historical GPU run bit for bit.

The frozen binary label is the endpoint for empirical tables in this draft. The current grader is tested independently, but passing its regression cases is not evidence that every historical label matches its current behavior or a human adjudication. Changes to the grader require a new label census and versioned results. Chapter 6 explains the consequence for scientific claims and final archival work.

1.6 Thesis organization

Chapter 2 defines the stochastic model, proves the general stopping result, and establishes the limits of drift-sign and uncertainty arguments. Chapter 3 documents the corpora, generation contracts, instrumentation, grading, grouping, and reproducibility procedures. Chapter 4 presents the measured boundary and detector evidence,

including falsification results. Chapter 5 describes the online controller, its runtime checks, and the measured accuracy-cost trade-offs. Chapter 6 discusses generalization, interpretation, and the remaining steps needed for a final defended thesis. The appendices supply the reproducibility commands, claim-to-artifact map, and a record of requirements that cannot be established by code execution alone.

1.7 Related work and the thesis position

Chain-of-thought prompting studies how explicit intermediate reasoning can improve language-model performance [1]. Its benefit does not imply that every additional reasoning increment is valuable. The stopping problem asks a conditional continuation question after a particular prefix, rather than comparing a reasoning prompt with a direct-answer prompt. The two questions can be complementary: intermediate work creates opportunities for repair, while a stopping policy decides how long to continue the chosen protocol.

Verifier training uses examples of completed solutions to distinguish useful candidates [2]. Process supervision instead supplies feedback on intermediate steps [6]. The present correctness target is the answer at a saved response boundary. It does not certify that each preceding mathematical statement is valid. Consequently, an answer-level detector in this project is not automatically a process verifier, and a high ranking score on saved responses is not a certificate of all reasoning leading to them.

Self-consistency samples several reasoning paths and aggregates their answers [7]. That idea motivates agreement features, but agreement and stopping have different cost structures. An aggregate answer can improve after additional paths are generated, while a stop requires deciding whether those generations are worth their cost. A peer

feature is therefore useful only within an explicit allocation and timing protocol. Treating thirteen completed peer trajectories as a free feature of one model would obscure the relevant computation comparison.

The overthinking literature studies excessive test-time reasoning and strategies for reducing it [8]. This thesis uses response-and-revision sequences and an explicit reward-minus-cost objective to make a related trade-off observable. It does not claim equivalence between its revision steps and every internal reasoning token used in other systems. The protocol-specific empirical distinction allows the results to be compared critically rather than assumed universal.

Optimal stopping provides the mathematical background for deciding between immediate reward and conditional future value [13], [14]. Its Bellman and Snell constructions are established theory. The contribution here is their careful application to hidden answer correctness, observable prefixes, computation accounting, and finite response horizons. In particular, the thesis distinguishes the general optimum from a myopic hazard approximation and supplies exact examples showing why the distinction matters.

Sequential uncertainty theory provides tools for controlling repeated statistical decisions under suitable sampling assumptions [15], [16]. Those tools motivate a principled eventual stopping certificate, but they do not validate an arbitrary confidence threshold or a reused bootstrap interval. The project therefore treats probability estimation and coverage as empirical and mathematical requirements that remain separate from causal implementation. This position makes the stopping controller inspectable while preserving the limits of the currently available evidence.

Chapter 2 Mathematical formulation and stopping theory

2.1 Probability model and admissible information

Fix a finite horizon N and a deterministic earliest permitted stop $m \in \{0, \dots, N\}$. Work on a probability space $(\Omega, \mathcal{A}, \mathbb{P})$. Let Y^* be the reference answer, H_t the information actually available after decision step t , and

$$\mathcal{F}_t = \sigma(H_0, \dots, H_t), \quad A_t = a_t(H_0, \dots, H_t), \quad C_t = \mathbf{1}\{A_t = Y^*\}.$$

All observations, executed peer calls, verifier outputs, and controller randomness used by a decision belong in this filtration. The runtime interface does not supply offline ground-truth fields, future revisions, future peer responses, or features computed using an unfinished trace. These are interface restrictions; mathematical measurability is governed by the information model and the caveat below. The correctness indicator need not be observable, even though the answer is.

The filtration is a statistical information model, not a model of computational hardness. If the reference answer is a known deterministic measurable function of a fully revealed task, then the selected candidate's correctness is $\mathcal{F}_t$ -measurable and $q_t = C_t$, even without a gold input field. All results below still hold in that degenerate case.

Nondegenerate conditional uncertainty requires a specified latent-reference generative model or an explicitly coarser filtration of decision statistics, with admissible policies restricted to that information. A fitted probability based on prefix summaries does not establish a posterior conditioned on all mathematical semantics of a task.

The candidate A_t means the answer that the specified policy would emit if it stopped at t . If the policy selects among current, earlier, or peer answers, define A_t by that causal selection procedure. Results for the current generator answer do not automatically apply to a different selected answer. If every available candidate in a finite, observable candidate set can be selected freely, the optimal immediate correctness reward is $\max_{a \in \mathcal{B}_t} \mathbb{P}(Y^* = a \mid \mathcal{F}_t)$, where $\mathcal{B}_t$ is the candidate set available at t .

A policy's stop time τ is admissible when

$$m \leq \tau \leq N, \quad \{\tau \leq t\} \in \mathcal{F}_t \quad \text{for every } t.$$

Let K_t be the nonnegative, nondecreasing, integrable, adapted cumulative cost of **all** computation executed by t , including peers and probes. Put

$$q_t = \mathbb{E}[C_t \mid \mathcal{F}_t], \quad G_t = q_t - K_t, \quad c_t = \mathbb{E}[K_{t+1} - K_t \mid \mathcal{F}_t].$$

The objective is $\sup_{\tau} \mathbb{E}[C_{\tau} - K_{\tau}]$. The step-cost specialization is $K_t = \lambda(t - m)$ for $t \geq m$; previously incurred constant costs do not change the optimizer. The empirical choice $\lambda = 0.05$ is a utility convention, not a universal monetary cost or an assertion about token savings.

Lemma 1 (observable reward reduction). For every admissible stop time,

$$\mathbb{E}[C_{\tau} - K_{\tau}] = \mathbb{E}[G_{\tau}].$$

Proof. The event $\{\tau = t\}$ belongs to $\mathcal{F}_t$, so $\mathbb{E}[\mathbf{1}_{\{\tau=t\}} C_t] = \mathbb{E}[\mathbf{1}_{\{\tau=t\}} q_t]$. Sum this equality over the finite set $t = m, \dots, N$, and subtract $\mathbb{E}[K_{\tau}]$. $\square$

For stopping-only comparisons, a full potential trace may be defined even after a policy would stop. Its prefixes must have the same law as the actual generation process up to

stopping. A controller that changes prompts, sampling, peer allocation, or answer selection changes the process; a retrospective evaluation must account for that change.

2.2 Exact binary correctness drift

For $t < N$, define the conditional joint transition masses

$$r_t = \mathbb{P}(C_t = 0, C_{t+1} = 1 \mid \mathcal{F}_t), \quad s_t = \mathbb{P}(C_t = 1, C_{t+1} = 0 \mid \mathcal{F}_t),$$

and the conditional repair and corruption probabilities

$$\alpha_t = \begin{cases} r_t/(1 - q_t), & q_t < 1, \\ 0, & q_t = 1, \end{cases} \quad \beta_t = \begin{cases} s_t/q_t, & q_t > 0, \\ 0, & q_t = 0. \end{cases}$$

The value assigned on a zero-probability conditioning state is arbitrary in $[0, 1]$: its multiplier is zero. These quantities are probabilities per decision transition, not continuous-time intensities. No Markov, independence, monotonicity, or observability of C_t is needed for the following identity.

Theorem 2 (conditional drift identity). Almost surely,

$$\mu_t := \mathbb{E}[G_{t+1} - G_t \mid \mathcal{F}_t] = (1 - q_t)\alpha_t - q_t\beta_t - c_t.$$

Proof. Pointwise, $C_{t+1} - C_t = \mathbf{1}_{\{C_t=0, C_{t+1}=1\}} - \mathbf{1}_{\{C_t=1, C_{t+1}=0\}}$. The tower property and nested filtrations give

$$\mathbb{E}[q_{t+1} \mid \mathcal{F}_t] = \mathbb{E}[\mathbb{E}[C_{t+1} \mid \mathcal{F}_{t+1}] \mid \mathcal{F}_t] = \mathbb{E}[C_{t+1} \mid \mathcal{F}_t].$$

Subtract $q_t = \mathbb{E}[C_t \mid \mathcal{F}_t]$, use the indicator identity, and subtract c_t . $\square$

Consequently, with $D_m = 0$ and $D_t = \sum_{s=m}^{t-1} \mu_s$, $M_t = G_t - G_m - D_t$ is an integrable martingale. This is a discrete Doob decomposition. The process q_t need not be a

martingale: it concerns the changing answer C_t , rather than the conditional probability of one fixed event.

For a sample of n observed transitions, let n_0, n_1 be the numbers of current incorrect and correct states and let n_{01}, n_{10} count repairs and corruptions. On that same sample,

$$\left(1 - \frac{n_1}{n}\right) \frac{n_{01}}{n_0} - \frac{n_1}{n} \frac{n_{10}}{n_1} = \frac{n_{01} - n_{10}}{n} = \frac{1}{n} \sum_i (C_{i,t+1} - C_{i,t}),$$

with the zero-denominator convention above. Frequencies n_{01}/n and n_{10}/n are already joint event frequencies; multiplying them again by $1 - q$ and q is incorrect. Matching this sample identity verifies arithmetic, not an online probability model. A product of separately averaged, instance-specific probabilities generally differs from the average of their products.

Corollary 2.1 (affine stakes). Suppose a correct answer earns $v \geq 0$ and an incorrect answer loses $p \geq 0$, with $v + p > 0$, and computation still costs K_t . The observable stop reward and its conditional drift become

$$G_t^{v,p} = vq_t - p(1 - q_t) - K_t = (v + p)q_t - p - K_t,$$

$$\mu_t^{v,p} = (v + p)((1 - q_t)\alpha_t - q_t\beta_t) - c_t.$$

Proof. The realized terminal reward is $vC_t - p(1 - C_t) - K_t$. Apply Lemma 1 and the linearity of conditional expectation to Theorem 2. $\square$

For constant computation cost, the correctness-versus-compute trade-off is therefore determined by cost divided by $v + p$; $-p$ is an additive constant. Increasing stakes can change the optimal policy. Under a fixed potential trace law and nondecreasing computation costs, optimal continuation regions expand as $v + p$ increases; Corollary

3.1 proves the corresponding ordering of earliest optimal stop times. That ordering need not hold for empirically thresholded policies, or when stakes change the generation process, answer selection, or costs. All comparisons must state both scales.

2.3 General finite-horizon optimal stopping

Theorem 3 (Bellman recursion and Snell envelope). Set

$$S_N = G_N, \quad S_t = \max\{G_t, \mathbb{E}[S_{t+1} \mid \mathcal{F}_t]\}, \quad t = N-1, \dots, m.$$

Then S is the smallest integrable supermartingale dominating G . For each $t \geq m$,

$$S_t = \operatorname{ess\,sup}_{\tau: t \leq \tau \leq N} \mathbb{E}[G_\tau \mid \mathcal{F}_t], \quad \tau_t^* = \inf\{s \in \{t, \dots, N\} : S_s = G_s\}$$

is an optimal stopping time, and $S_t = \mathbb{E}[G_{\tau_t^*} \mid \mathcal{F}_t]$.

Proof. Backward induction gives integrability, adaptation, $S_t \geq G_t$, and $S_t \geq \mathbb{E}[S_{t+1} \mid \mathcal{F}_t]$. If W is any integrable supermartingale with $W_s \geq G_s$, then $W_N \geq S_N$. If $W_{s+1} \geq S_{s+1}$, the supermartingale property and domination give $W_s \geq \max\{G_s, \mathbb{E}[S_{s+1} \mid \mathcal{F}_s]\} = S_s$. This proves minimality by induction.

For any bounded admissible τ , finite optional sampling, or the finite telescoping sum of conditional supermartingale increments, gives $\mathbb{E}[G_\tau \mid \mathcal{F}_t] \leq \mathbb{E}[S_\tau \mid \mathcal{F}_t] \leq S_t$. The first contact time is measurable and no later than N . On $\{s < \tau_t^*\}$, $S_s > G_s$, so the recursion forces $S_s = \mathbb{E}[S_{s+1} \mid \mathcal{F}_s]$. Thus $(S_{s \wedge \tau_t^*})_{s=t}^N$ is a martingale. At contact $S_{\tau_t^*} = G_{\tau_t^*}$, giving equality in the upper bound. Attainment establishes the essential-supremum representation. $\square$

This is the standard finite-horizon optimal stopping construction; see Ferguson, Chapter 3, Section 3.2. The proof here specializes it to the observable correctness-minus-cost

reward and includes the explicit protocol floor.

The optimal continuation advantage is

$$\Delta_t^* = \mathbb{E}[S_{t+1} \mid \mathcal{F}_t] - G_t = \mu_t + \mathbb{E}[S_{t+1} - G_{t+1} \mid \mathcal{F}_t] \geq \mu_t.$$

The second term is the value of future choices. Therefore $\mu_t > 0$ is sufficient to favor continuation at $t < N$; $\mu_t \leq 0$ alone is insufficient to justify stopping. At $N - 1$ the terms coincide. The optimal policy stops at the first $\Delta_t^* \leq 0$, or at N .

Corollary 3.1 (stake monotonicity under a fixed process). Compare affine stakes with $0 < w_1 \leq w_2$, where $w_j = v_j + p_j$. Suppose that the filtration, candidate answers, potential trace law, horizon, floor, and nondecreasing cumulative cost process are identical in both problems. Then their earliest optimal contact times satisfy $\tau^*(w_1) \leq \tau^*(w_2)$ almost surely.

Proof. For $t < N$, the normalized advantage of being required to continue at least once is

$$\frac{\Delta_t^*(w)}{w} = \operatorname{ess\,sup}_{\tau: t+1 \leq \tau \leq N} \mathbb{E} \left[q_\tau - q_t - \frac{K_\tau - K_t}{w} \mid \mathcal{F}_t \right].$$

The constant incorrect-answer offset cancels. Since $K_\tau - K_t \geq 0$, each conditional expectation is nondecreasing in w , as is its essential supremum. Consequently $\Delta_t^*(w_1) > 0$ implies $\Delta_t^*(w_2) > 0$. Along any fixed path, the larger-stakes policy cannot stop while the smaller-stakes policy has continued at every decision so far: each of those strict continuation conditions also holds for larger stakes. With the same tie convention and forced terminal stop, this proves the stated ordering. $\square$

This result does not assert that accuracy itself increases strictly, that all fitted drift boundaries obey the ordering, or that a negative drift becomes positive merely because stakes increase.

2.4 When a drift-sign boundary is optimal

Theorem 4 (persistent nonpositive drift). Define

$$T_c = \min(\{t \in \{m, \dots, N-1\} : \mu_t \leq 0\} \cup \{N\}).$$

Assume, almost surely, that $\mu_s \leq 0$ for every $s \in \{T_c, \dots, N-1\}$. Then T_c maximizes $\mathbb{E}[G_\tau]$ over all admissible stop times.

Proof. For any such τ , telescoping and the measurability of $\{\tau > s\}$ yield

$$\mathbb{E}[G_\tau] = \mathbb{E}[G_m] + \mathbb{E} \sum_{s=m}^{N-1} \mathbf{1}_{\{\tau > s\}} \mu_s.$$

For $s < T_c$, $\mu_s > 0$ and $\mathbf{1}_{\{T_c > s\}} = 1$. For $s \geq T_c$, $\mu_s \leq 0$ and $\mathbf{1}_{\{T_c > s\}} = 0$. Accordingly,

$$(\mathbf{1}_{\{T_c > s\}} - \mathbf{1}_{\{\tau > s\}}) \mu_s \geq 0$$

on every path. Sum, take expectations, and apply the telescoping identity to T_c . $\square$

This is a condition on the **conditional drifts along sample paths**. A single crossing of an empirical population mean curve does not establish it. Theorem 4 is a special case of monotone stopping problems, whose one-stage look-ahead conditions are discussed in Ferguson, Chapter 5.

Corollary 4.1 (a sufficient structural condition). Suppose cost is constant $c_t = \lambda$, and, almost surely along every path, $q_{t+1} \geq q_t$, $\alpha_{t+1} \leq \alpha_t$, and $\beta_{t+1} \geq \beta_t$ for $m \leq t <$

$N - 1$, for specified versions of the hazards. Then μ_t is nonincreasing and Theorem 4 applies.

Proof. Subtract adjacent drift expressions:

$$\begin{aligned}\mu_{t+1} - \mu_t &= (1 - q_{t+1})(\alpha_{t+1} - \alpha_t) + \alpha_t(q_t - q_{t+1}) \\ &\quad - q_{t+1}(\beta_{t+1} - \beta_t) - \beta_t(q_{t+1} - q_t).\end{aligned}$$

Every term is nonpositive because all probabilities lie in $[0, 1]$. With variable predictable cost an extra term $-(c_{t+1} - c_t)$ appears; nondecreasing c_t is an additional sufficient condition. $\square$

These sufficient assumptions are strong. They are not established by a high correctness AUC, a good average utility curve, or a monotone fitted hazard curve. A noisily estimated posterior may increase and decrease after new observations even when the marginal accuracy improves.

Exact counterexample 1: delayed repair

Use times 0, 1, 2, constant cost $1/10$, and correctness probabilities $q = (0, 0, 1)$. An exact-answer realization has a latent binary reference Y^* , a surely incorrect placeholder at times 0 and 1, and a generator that outputs Y^* at time 2. No information at time 0 or 1 reveals which reference answer will be returned. Then $G = (0, -1/10, 4/5)$ and $\mu_0 = -1/10$. The myopic boundary stops at 0 with reward 0. The optimal stop is 2 with reward $4/5$. A second system with $q = (0, 0, 0)$ has the same current $(q_0, \alpha_0, \beta_0, c_0) = (0, 0, 0, 1/10)$ but optimally stops at 0. Thus even the exact current hazard triple cannot identify a general multi-step optimal policy. The example can be shifted to a floor $m = 2$ without changing the comparison.

Exact counterexample 2: a population curve misses adaptive value

Take $q_0 = 1/2$ and cost $1/10$. At time 1, a fair observed signal gives branch A with $q_1 = 2/5$ or branch B with $q_1 = 3/5$. At time 2, set $q_2 = 1$ on A and $q_2 = 0$ on B. Both unconditional one-step drifts equal $-1/10$, and the deterministic-time expected rewards are $1/2, 2/5, 3/10$. Yet continuing once, then continuing on A and stopping on B, achieves

$$\frac{1}{2}(1 - 2/10) + \frac{1}{2}(3/5 - 1/10) = 13/20 > 1/2.$$

For an exact-answer realization, let $Y^* \in \{0, 1\}$ have prior probability $1/2$ of 1, and use candidate answer 1 at times 0 and 1. Set $\mathbb{P}(A, Y^* = 1) = 1/5$, $\mathbb{P}(A, Y^* = 0) = 3/10$, $\mathbb{P}(B, Y^* = 1) = 3/10$, and $\mathbb{P}(B, Y^* = 0) = 1/5$. At time 2 the generator returns Y^* on branch A and a surely incorrect placeholder on branch B. The controller uses only the observed branch to make its time-1 decision. Thus the examples respect a common reference answer and use no hindsight information to make decisions.

2.5 Partial observation and feature models

For a causal feature vector $Z_t = \phi_t(H_0, \dots, H_t)$, define

$$\tilde{q}_t = \mathbb{P}(C_t = 1 \mid Z_t), \quad \tilde{\alpha}_t = \mathbb{P}(C_{t+1} = 1 \mid C_t = 0, Z_t), \quad \tilde{\beta}_t = \mathbb{P}(C_{t+1} = 0 \mid C_t = 1, Z_t).$$

Their hazard expression equals $\mathbb{E}[C_{t+1} - C_t \mid Z_t]$ exactly, subject to the same null-state convention. This feature-conditional quantity need not equal the history-conditional drift in Theorem 2. In particular, unrelated $\sigma(Z_t)$ at different times need not form a filtration.

A Bellman equation on a compressed state requires sufficiency: conditional on that state, the law of future observations, reward, and cost must not depend on omitted history. Alternatively a correctly specified partially observed Markov model can use the

full posterior over its latent state as the belief state. A scalar correctness posterior and two current hazards do not by themselves provide that transition model. Without such assumptions, a fitted hazard rule is an empirical myopic approximation.

Retrospective correctness classifiers, continuation predictors, and optimal stopping policies are different estimands. Training with labels is allowed; using those labels or future observations to decide on the same live instance is not. A prefix-stable feature transform must be fitted before evaluation or updated solely from information available by that prefix. Full-family or full-trace normalizations may change earlier decisions when future rows are added, even without directly using correctness labels.

2.6 Calibration and perturbation bounds

Discrimination is not probability accuracy. AUC is unchanged by a strictly increasing transform, while the drift sign generally changes. Even marginal calibration $\mathbb{E}[C_t \mid \hat{q}_t] = \hat{q}_t$ does not imply $\hat{q}_t = \mathbb{E}[C_t \mid \mathcal{F}_t]$: a constant $1/2$ score is marginally calibrated in a balanced population whose available history perfectly separates correct and incorrect cases.

Proposition 5 (class weighting changes the population target). For a binary target with conditional probability $p \in (0, 1)$, positive class weights w_1, w_0 , and an unconstrained probability prediction r , the weighted Bernoulli log loss is uniquely minimized at

$$r = \frac{w_1 p}{w_1 p + w_0(1 - p)}, \quad p = \frac{w_0 r}{w_1(1 - r) + w_0 r}.$$

Proof. Differentiate $-w_1 p \log r - w_0(1 - p) \log(1 - r)$; setting the derivative to zero yields the first formula. Strict positivity of the second derivative gives uniqueness.

Algebra gives the inverse. $\square$

`equation_analysis.py` and `trace_analysis.py` fit class-balanced classifiers. Thus their uncorrected probability outputs are not guaranteed to be target-distribution q, α, β , even if ranking is useful. For example, balanced weighting in a population with prevalence $1/10$ maps its constant posterior to $1/2$. With $\alpha = 1/50, \beta = 1/200, \lambda = 1/100$, changing only q from $1/10$ to $1/2$ changes drift from $3/400 > 0$ to $-1/400 < 0$. The inverse formula describes a population optimum; model misspecification, regularization, estimated weights, and distribution shift still require independent probability validation. ECE and Brier scores are useful diagnostics but do not provide uniform conditional error guarantees.

Proposition 6 (drift perturbation and interval propagation). If $|\hat{q} - q| \leq \varepsilon_q$, $|\hat{\alpha} - \alpha| \leq \varepsilon_\alpha$, $|\hat{\beta} - \beta| \leq \varepsilon_\beta$, and $|\hat{c} - c| \leq \varepsilon_c$, with all probabilities in $[0, 1]$, then

$$|\hat{\mu} - \mu| \leq 2\varepsilon_q + (1 - \hat{q})\varepsilon_\alpha + \hat{q}\varepsilon_\beta + \varepsilon_c.$$

If simultaneous valid intervals are $q \in [q_L, q_U]$, $\alpha \in [a_L, a_U]$, $\beta \in [b_L, b_U]$, and $c \in [c_L, c_U]$, their exact rectangular drift bounds, provided the three probability intervals are contained in $[0, 1]$, are

$$L = (1 - q_U)a_L - q_U b_U - c_U, \quad U = (1 - q_L)a_U - q_L b_L - c_L.$$

Proof. Subtract the true expression from the fitted expression as

$$\hat{\mu} - \mu = (1 - \hat{q})(\hat{\alpha} - \alpha) - \hat{q}(\hat{\beta} - \beta) - (\hat{q} - q)(\alpha + \beta) - (\hat{c} - c).$$

Apply the triangle inequality and $\alpha + \beta \leq 2$. The drift is increasing in α and decreasing in q, β, c on their domains, so the stated corners attain its minimum and maximum on the rectangle. $\square$

This propagation result assumes genuine interval coverage of the relevant conditional probabilities. It cannot manufacture such intervals from AUC, a chosen safety margin, cross-validation alone, or calibration plots.

2.7 What sequential uncertainty control can certify

Proposition 7 (a precisely scoped first-crossing guarantee). Suppose U_t is observable at decision t and

$$\mathbb{P}(\exists t \in \{m, \dots, N-1\} : \mu_t > U_t) \leq \delta.$$

Let τ_U be the first $U_t \leq 0$ after the floor, with fallback N . Then

$$\mathbb{P}(\tau_U < N \text{ and } \mu_{\tau_U} > 0) \leq \delta.$$

Proof. On the simultaneous-coverage event, $\mu_{\tau_U} \leq U_{\tau_U} \leq 0$ whenever $\tau_U < N$. Every violation is therefore contained in the coverage-failure event. $\square$

This certifies a nonpositive **one-step** drift, not optimality, retained accuracy, or proximity to a hindsight oracle. A corresponding simultaneous upper bound on Δ_t^* certifies the Bellman stop condition. To interpret a bound on μ_t as a statement about the myopic boundary one must separately impose the relevant structural assumptions.

For a probe construction, distinguish the pre-probe sigma-field $\mathcal{H}_t$ from the post-probe decision information $\mathcal{F}_t$. Let θ_t be an $\mathcal{H}_t$ -measurable target, and assume the n_t probes $X_{t,i} \in [a, b]$ are conditionally independent given $\mathcal{H}_t$, each with conditional mean θ_t . Fix their count before observing them. Conditional Hoeffding gives a bound for θ_t , not automatically for μ_t :

$$U_t^{\text{pre}} = \bar{X}_t + (b - a) \sqrt{\frac{\log(1/\delta_t)}{2n_t}}.$$

Taking $\delta_t = \delta/[r(r+1)]$ for decision index $r = t - m + 1 \geq 1$ gives $\sum_t \delta_t \leq \delta$, so a union bound establishes simultaneous coverage of the θ_t targets. Decisions can depend on earlier data; independence between different decision times is unnecessary. Within-prefix conditional independence and the correct conditional mean are essential. Adaptively choosing the number of probes requires a bound also uniform in probe count, or a further failure-budget allocation. The bounded-sample result is due to Hoeffding (1963).

Probe observations and their cost must enter the actual decision filtration. If they reveal information that changes the live conditional continuation gain, a bound for the pre-probe conditioning target is not automatically a bound for the post-probe target. Proposition 7 follows from this construction only if $\theta_t = \mu_t$ after the probes, or if an additional justified comparison supplies a bound for the post-probe μ_t . Shared unknown ground truth can make apparently independent generations dependent after conditioning only on the visible prefix. Verified probe labels are also not generally available at deployment. An empirical-Bernstein formula requires its own theorem, variance convention, and sampling assumptions; merely inserting a sample variance does not prove those conditions. The modern confidence sequence framework is developed by Howard et al. (2021).

For completeness, a valid elementary e-process can be obtained when $X_i \in [-1, 1]$ is adapted to a sampling filtration $\mathcal{G}_i$ and the null satisfies $\mathbb{E}[X_i \mid \mathcal{G}_{i-1}] \geq 0$ for every i . For any fixed $\eta \in [0, 1)$,

$$E_n^{(\eta)} = \prod_{i=1}^n (1 - \eta X_i)$$

is a nonnegative supermartingale starting at 1: its next conditional multiplier has expectation at most 1. A fixed convex mixture of such processes has the same property. Finite optional sampling at the first threshold crossing, followed by monotone limits, gives

$$\mathbb{P}(\sup_n E_n \geq 1/\delta) \leq \delta.$$

This rejects that conditional-mean null for the specified sampling process. It does not estimate a different instance's present continuation gain. If $X_i = B$ for all i , with one shared fair sign B , all marginal means are zero, yet $(1 - X_i/2)$'s ten-factor product exceeds 20 on the negative branch with probability $1/2$. Hence marginal means and boundedness alone do not justify a 5% sequential guarantee.

In `trace_analysis.py`, empirical-Bernstein bounds and mixture products are computed across labeled task rows within a step, then used for a common step-level diagnostic. That construction is not an observable per-instance confidence bound. Repeated runs of the same task also need dependence handling for population inference. This note makes no claim that the repository's raw fitted probabilities or historical detectors satisfy Proposition 7's coverage premise.

2.8 Scope of the empirical claims

Mathematical object	Required information or assumptions	Defensible interpretation
Exact conditional hazard identity	Binary reward, correct conditioning, common transition sample	Algebraic one-step decomposition
Snell/Bellman policy	Conditional future law and all computation costs	Optimal finite-horizon causal stop
First nonpositive true drift	Persistent nonpositive drift after first crossing	Optimal under Theorem 4
Fitted first nonpositive drift	Causal features and validated probability estimates	Empirical myopic stopping rule
First nonpositive upper bound	Simultaneous coverage of the stated target	Probabilistic one-step sign certificate
Hindsight best recorded step	Completed labels and trace	Infeasible upper benchmark
Correctness AUC	Held-out target labels and scores	Ranking quality on that population

A hindsight maximum satisfies $\mathbb{E}[\max_{m \leq t \leq N}(C_t - K_t)] \geq \sup_{\tau} \mathbb{E}[C_{\tau} - K_{\tau}]$, because it dominates every realized admissible stop reward. Its maximizing time generally fails the stopping-time condition. Therefore a hindsight oracle gap includes unavailable information, and a reported early/late stop relative to that oracle is not a formal Type I error rate.

A minimum-step floor is a protocol constraint. It restricts the policy class and can improve a particular fitted policy empirically; it cannot improve the unconstrained optimum, because that optimum already includes every policy respecting the floor. For positive constant step cost, q_t measures accuracy while $q_t - \lambda(t - m)$ measures utility; their peaks need not coincide. Any savings claim must compare actual executed

tokens/calls with a defined baseline, not convert stopped step indices into hardware savings without measurement.

The results are relative to the correctness target C_t . A passed grading test suite does not prove every empirical label correct, and an exact-answer benchmark theorem is not a theorem about open-ended semantic correctness. The standard optimal stopping machinery here is not claimed as a novel theorem or a universal law across model architectures.

Chapter 3 Experimental setup and methods

3.1 Corpus separation

The project contains two main collections with different purposes. The canonical model-domain matrix is a variable-horizon response-and-revision collection used for boundary and controlled policy analyses. Its current analysis describes 75,965 sanitized trajectories and 798,770 raw saved step rows over 52 model-domain cells. The word sanitized matters: raw malformed fragments and inconsistent records are handled by the analysis pipeline, so a raw row count and an eligible trajectory count are different denominators. The standardized detector collection contains 144,440 rows from 28,888 five-step trajectories and 2,948 unique task identifiers. Its 52 cells support detector training and task-grouped scoring.

The collections overlap in scientific subject matter and include repeated benchmark questions. They are not independent replications and must not be summed into a single sample size. The earlier replay experiment also reuses a 1,500-trajectory Qwen2.5-7B/GSM8K cell from existing evidence. Its replayed decisions do not create new language-model generations.

Table 1. Standardized five-step corpus.

Domain	Effective split	Tasks	Trajectories	Rows
GSM8K	train	1,000	8,064	40,320
MATH-500	test	500	6,500	32,500
ARC-Challenge	test	1,000	8,500	42,500
GPQA main	train (inferred; request test)	448	5,824	29,120
Total	four domains	2,948	28,888	144,440

The standardized task count exceeds the canonical matrix's task count because the collection uses different benchmark sample allocations. Each standardized trajectory has five rows, so 28,888 multiplied by five gives exactly 144,440. Multiplying the unique task count by thirteen and five would be incorrect because model-domain cells have unequal sample sizes. Counts are derived from source-qualified trajectory keys, not from that shortcut.

3.2 Model roster

The canonical matrix includes thirteen model configurations: DeepSeek-R1-Distill-Qwen-1.5B and 7B; Qwen2.5-0.5B, 3B, 7B, 14B, and 32B Instruct; InternLM3-8B-Instruct; Llama-3.1-8B-Instruct; Mistral-7B-Instruct-v0.3; Mistral-Small-Instruct-2409; Phi-4-mini-instruct; and Yi-1.5-9B-Chat. The standardized detector collection replaces InternLM3 with Qwen3.5-9B. These are distinct rosters, even though both have thirteen entries.

Table 2. Model configurations. Source: cell metadata.

Model ID	Recorded scale	Membership
deepseek-ai/DeepSeek-R1-Distill-Qwen-1.5B	1.5B	Matrix + detector
deepseek-ai/DeepSeek-R1-Distill-Qwen-7B	7B	Matrix + detector
internlm/InternLM3-8B-Instruct	8B	Matrix only
meta-llama/Llama-3.1-8B-Instruct	8B	Matrix + detector
mistralai/Mistral-7B-Instruct-v0.3	7B	Matrix + detector
mistralai/Mistral-Small-Instruct-2409	22B	Matrix + detector
microsoft/Phi-4-mini-instruct	4B	Matrix + detector
Qwen/Qwen2.5-0.5B-Instruct	0.5B	Matrix + detector
Qwen/Qwen2.5-14B-Instruct	14B	Matrix + detector
Qwen/Qwen2.5-32B-Instruct	32B	Matrix + detector
Qwen/Qwen2.5-3B-Instruct	3B	Matrix + detector
Qwen/Qwen2.5-7B-Instruct	7B	Matrix + detector
01-ai/Yi-1.5-9B-Chat	9B	Matrix + detector
Qwen/Qwen3.5-9B	9B	Detector only

The historical alias `mistral_small_24b_2409` identifies the 2409 model, whose recorded specification is 22B. An alias is not a reliable parameter count. Scientific tables use the documented model specification and retain the alias only where needed to locate source files. Likewise, a speculative entry in a software catalog is not evidence that a model was included in a completed experiment; the actual manifests and trace metadata determine inclusion.

The model panel allows descriptive comparisons of families and scale. It does not randomize architecture, pretraining data, instruction tuning, distillation, or tokenization. A difference between two model families therefore cannot be attributed uniquely to scale. Within the Qwen ladder, common family identity reduces some confounding, but still does not create a controlled intervention on parameter count alone.

3.3 Benchmark tasks and splits

GSM8K contains grade-school mathematical word problems and was introduced in the verifier study of Cobbe and colleagues [2]. MATH provides competition-style mathematical problems [3]. ARC-Challenge provides difficult multiple-choice science questions [4]. GPQA provides graduate-level multiple-choice scientific questions [5]. These references identify the benchmark tasks; they do not validate this project's chosen subsets or grading implementation.

The canonical experiment registry specifies GSM8K train, MATH test, ARC-Challenge test, and GPQA main train. The standardized collection also uses GSM8K train, MATH test, and ARC-Challenge test. Its GPQA metadata records `test`, but the current loader unconditionally requests the `train` split; the saved identifiers use `gpqa_main`. The requested and inferred effective split are retained as a provenance discrepancy rather than silently equated. The historical loader revision is not recorded, so this effective-split interpretation uses the current implementation and stored task identities. Thus a dataset's published split name and a detector's held-out fold mean different things. A detector can be evaluated on a task-held-out fold drawn from a benchmark training split. That is an internal task holdout, not an official benchmark test-set evaluation.

SVAMP does not appear in the authoritative standardized tournament manifest. It is excluded from the reported four-domain totals. The complete prompt, task identifier, reference answer, split, and any deterministic choice shuffle must remain associated with a question. An MCQ label such as B only has meaning relative to the exact displayed ordering. The project stores or reconstructs that ordering before evaluating the option label.

3.4 Response and revision protocol

A saved step is a complete response generated under the collection's prompt contract. For the canonical registry, seed 7 and dataset shuffle seed 17 are recorded. The temperature panel uses 0.1, 0.6, and 1.0, and the common completion-token cap is 256. Response horizons are domain-specific: ten increments for GSM8K and GPQA, fourteen for MATH, and eight for ARC. Individual metadata remains authoritative when a cell records a deviation or recovery event.

The standardized detector corpus has five saved increments per trajectory. Its generation settings include temperature 0.6 and seed 7. A repeated call with a revision prompt is not equivalent to revealing another fragment of one uninterrupted chain of thought. The revision prompt itself can change the distribution of the next answer. The stopping target is therefore defined relative to this complete sequential protocol, including prompt construction and sampling.

The minimum admissible stopping step is two in the principal experiments. This floor is a design choice informed by development analyses. It is not a theorem that every question requires a revision. Some trajectories are correct only at step one, and the floor can exclude their best candidate. Allowing an answer-selection policy to return a

previously saved candidate would define a different action space; such a policy must be compared separately.

At every decision, computation already incurred is sunk. The continuation decision should charge the incremental cost of future generation, not charge the prefix again. At evaluation, however, total policy cost includes every generated response up to the selected step. The distinction keeps the Bellman recursion and empirical cost accounting consistent.

3.5 Recorded signals and grading

Recorded signals can include candidate text, normalized answer, correctness, completion-token count, generation runtime, answer changes, log-probability summaries, entropy, hidden-state displacement, and peer agreement. Availability varies across experiments. A missing signal should be recorded as missing or excluded by a documented rule; treating it as a meaningful zero can alter a learned boundary.

Correctness labels use the benchmark's answer type. Numeric tasks use extraction and normalization; symbolic MATH tasks use the project's mathematical equivalence checks; multiple-choice tasks compare the parsed option with the exact displayed reference. Normalization must preserve information that matters to equivalence. For example, reducing an arbitrary plane equation to its right-hand side would identify distinct mathematical objects, and treating the phrase 'the third option' as the fraction one third would corrupt an MCQ answer.

The grader regression script checks thirty specific cases. The minimum-boundary script checks six cases. Their execution is recorded by command, because historical standalone scripts are not necessarily thirty and six discoverable pytest functions.

These checks defend the tested semantics. They do not constitute exhaustive validation of every mathematical answer or all historical parser versions.

All empirical labels used in the primary boundary tables are the stored correct values. The data-freeze tools do not mutate them. A future regrade must preserve the previous labels, record the grader version, compare changes by domain and answer type, and publish derived results under a new version. That procedure avoids silently changing the estimand halfway through a comparison.

3.6 Statistical units and grouping

Rows from one trajectory are dependent, and trajectories from the same task share the question and reference. Task-held-out detector folds keep all rows for a question outside the detector's training data. A source-qualified key combines the cell with its run identifier to prevent accidental merging of trajectories from different model-domain cells. Checking raw identifier collisions is useful, but source qualification remains the defined identity rule.

The strongest stored task-grouped detector analyses use five outer folds. For meta-model stacking, every upstream fitted component must also respect the outer split: its parameters and any calibrated thresholds must be fit using outer-training data alone. Producing a base score out of fold somewhere in the development pipeline does not automatically make a later outer-fold stacked result nested. Chapter 4 explains why the historical 0.955156 result falls short of that standard.

Cluster bootstraps resample questions when the same question appears in several trajectories. Model-domain estimator comparisons use the recorded cell bootstrap for their reported interval. These intervals answer different questions. A question bootstrap

estimates variability associated with the observed question population conditional on the fixed model panel; a cell bootstrap summarizes variability across the existing panel. Neither one is a multi-seed generation replication.

3.7 Outcomes and computation measures

Accuracy is the mean binary correctness label at the policy's selected answer. Historical per-trajectory step utility is $U_i(\tau_i) = C_{i,\tau_i} - 0.05(\tau_i - 1)$, so the mandatory first response is a common baseline and each additional increment incurs the penalty. This penalty is a utility convention, not five percent of physical power consumption or a universal economic price. Token utility is $C_{i,\tau_i} - 0.0002 \sum_{s=1}^{\tau_i} L_{i,s}$, where $L_{i,s}$ is measured completion tokens; 250 completion tokens carry the same penalty as one response increment. Charging every generated token includes the first response, which cancels in paired policy differences on the same trace. The theoretical reward in Chapter 2 can subtract the common prefix cost without changing the optimal continuation decision.

Token savings are one minus the ratio of total stopped-policy completion tokens to total full-horizon completion tokens, calculated on a common task panel. The ratio of totals differs from the average per-question saving and weights long generations more heavily. Both the definition and denominator must accompany a percentage. Prompt tokens, scoring passes, peer generations, controller inference, and runtime may have separate costs and cannot be omitted from a claim about total compute savings.

ROC-AUC measures ranking, with half credit for tied scores. Brier loss measures squared error of probabilistic predictions. Neither alone establishes a stopping guarantee. Class-balanced raw probability outputs generally target a reweighted posterior, as derived in Chapter 2; their held-out ranking or marginal calibration does not

validate natural-distribution conditional hazards. A detector with high pooled AUC can be poorly calibrated around the selected stopping threshold or fail on an underrepresented domain. The evaluation therefore includes micro, task-macro, domain-macro, worst-domain, and policy utility summaries.

3.8 Reproducibility and evidence freeze

The freeze records exact local byte hashes and canonical LF content hashes for the selected tournament files, cross-checks historical fingerprints, and pins supporting boundary and analysis artifacts. It also records source-code identities. File size and row count are checked in addition to content identity, because a compact summary alone can obscure selection errors. The reproducibility appendix gives the verification commands.

The workstation lock and recorded historical environment are separate evidence. Stored Blackwell metadata records a CUDA 13.0 runtime and PyTorch 2.13.0+cu130, whereas the current workstation has different versions. Compiler libraries, drivers, hardware, random-state handling, and nondeterministic kernels can affect a repetition even with an exact package list. The freeze supports inspection and re-analysis of the stored corpus; bit-identical regeneration is not established.

Every newly produced table and figure includes or is associated with its source paths and a hash manifest. The thesis build uses the recomputed evidence directory, not hand-edited numerical summaries. The separation allows a reader to inspect a claim, find the table that supplies it, and then verify the underlying frozen sources without reconstructing the author's conversational history.

Chapter 4 Empirical evidence of overthinking

4.1 Population transitions and net value

The boundary analysis measures whether another response increment improves average correctness enough to pay its cost. For a fixed panel, the next-step change in accuracy equals repair frequency minus corruption frequency. Subtracting the step penalty produces the empirical net gain. This is an exact decomposition of observed labels on a common transition-eligible panel, not an assumption about the model's internal reasoning.

Table 3. GSM8K transition panel; 19,500 trajectories and 500 task clusters per row.

Step	Accuracy	Repair count	Repair prob.	Corruption count	Corruption prob.	Net gain	95% interval
2	0.2401	3,145/14,818	0.2122	1,170/4,682	0.2499	+0.0513	[+0.0433, +0.0593]
4	0.4020	1,830/11,661	0.1569	1,102/7,839	0.1406	-0.0127	[-0.0186, -0.0065]

For the pooled GSM8K panel, step two has accuracy 0.2401. There are 3,145 repairs among 14,818 currently incorrect candidates and 1,170 corruptions among 4,682 currently correct candidates. The estimated repair and corruption probabilities are 0.2122 and 0.2499, respectively. Despite the larger conditional corruption probability, repairs outnumber corruptions because many more candidates are currently wrong. The net next-step gain after the 0.05 penalty is positive, approximately 0.0513.

At step four, the panel contains 1,830 repairs and 1,102 corruptions among 19,500 transition-eligible trajectories. Their difference increases accuracy by about 0.0373, which is less than the 0.05 penalty. Net gain is therefore approximately -0.0127. This example demonstrates why a negative net gain does not mean accuracy must fall: accuracy can still improve while the value of improvement is below its assigned cost.

Each of these two panels comprises 500 task clusters and 19,500 trajectories. The reported task-bootstrap intervals condition on the observed model and temperature panel. They characterize uncertainty in a population transition contrast, rather than a probability statement that a particular current answer is correct.

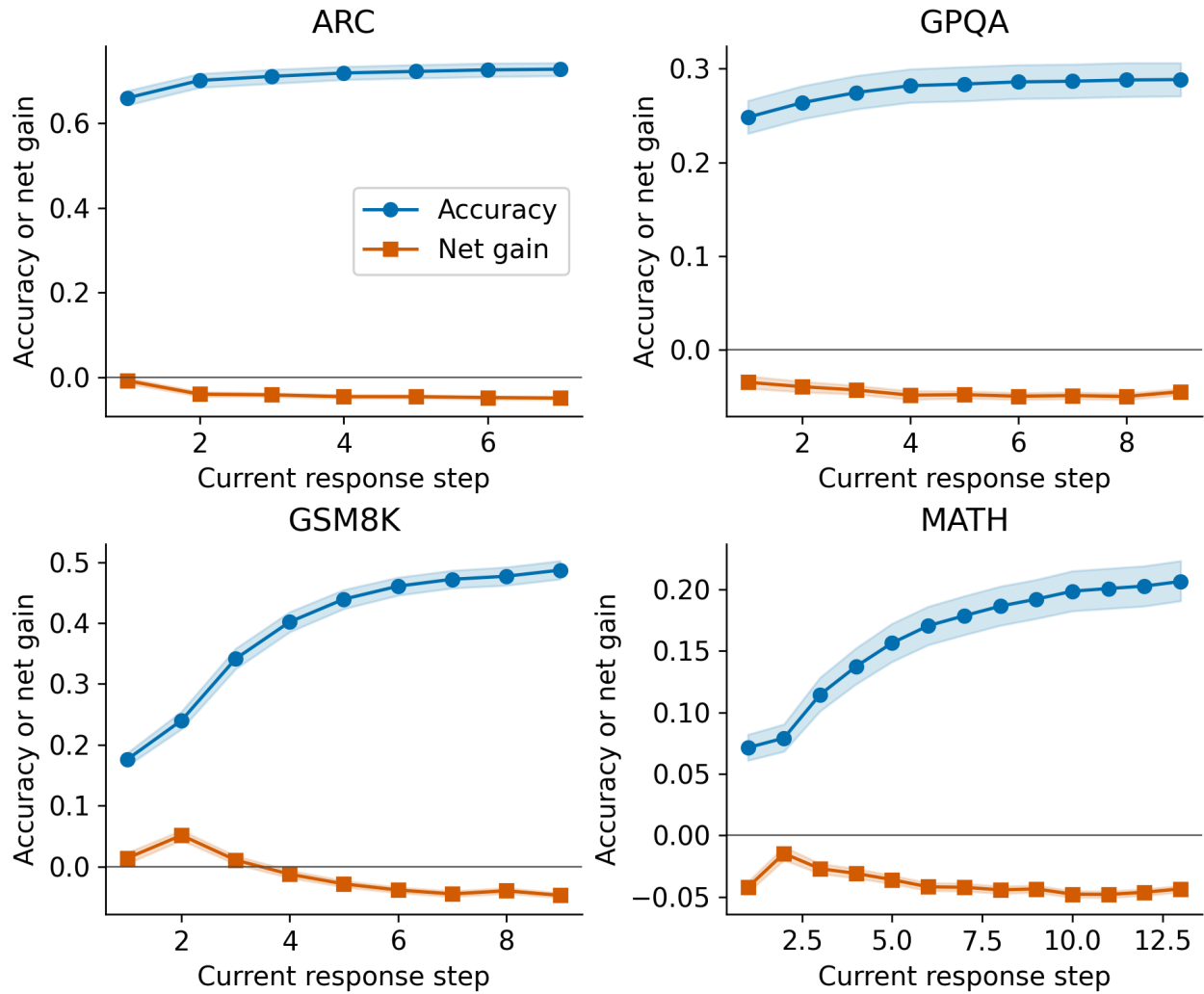


Figure 1. Population accuracy and net gain. Bands are task-cluster bootstrap intervals. The horizontal line marks zero gain, not zero accuracy.

4.2 Model and domain differences

Selected cells show different empirical crossings. In Qwen2.5-7B/GSM8K, net gain at step four is positive, approximately 0.0193, and step five is negative, approximately -0.0253. In Qwen2.5-32B/MATH, step five is positive, approximately 0.0187, and step

six is negative, approximately -0.0133. Each cell contains 1,500 trajectories over 500 questions. The intervals in the evidence tables quantify variation over those questions.

These results support a model- and domain-dependent continuation window. They do not support a universal claim that accuracy peaks at step two or three across the project. Nor do they prove that the first negative empirical gain is the globally best stopping point. The recorded curves can return to positive gain later. Selecting a final positive-to-negative crossing after seeing the complete curve is a retrospective descriptive choice, not a live stopping time.

The scale comparison also needs careful language. A larger model can have a longer useful revision window in a particular domain, but the design does not isolate parameter count from all architecture and training differences. The observed ladder is evidence about these specific models, settings, and questions. A universal scaling law would require additional models, independent generations, and a prespecified functional relationship tested on new data.

4.3 Controlled estimator comparisons

Table 4. Matched effects. Estimator intervals resample 52 cells; systems intervals resample 500 tasks.

Arm	Units	Matched effect	95% interval	Endpoint
N1 LOCO	75,965	+0.01210	[+0.00522, +0.01907]	step utility per trajectory
N1 LOMO	75,965	+0.01286	[+0.00683, +0.01911]	step utility per trajectory
N2a	75,965	-0.05694	[-0.08716, -0.03145]	step utility per trajectory
N2b	75,965	-0.06165	[-0.09168, -0.03547]	step utility per trajectory
N2c	75,965	+0.00331	[+0.00062, +0.00633]	step utility per trajectory
N3	75,965	+0.00781	[+0.00276, +0.01355]	step utility per trajectory
N4	75,965	+0.00210	[+0.00049, +0.00403]	step utility per trajectory
N5	1,500	+0.00000	[+0.00000, +0.00000]	loss-risk difference
N6	1,500	+0.14267	[+0.11133, +0.17533]	accuracy difference

Empirical-Bayes step hazards improve mean step utility by 0.00781 per trajectory relative to the matched cell-local logistic baseline, with a recorded 52-cell interval approximately [0.00276, 0.01355]. Lagged logistic features improve utility by approximately 0.00331, and the step-two churn threshold contrast improves it by approximately 0.00210. These are mean controlled contrasts, not hundreds of percentage points of accuracy.

The distinction between a controlled effect and an aggregate score matters. The historical '+593.55' for the pooled-hazard arm is a sum of utility differences over 75,965 trajectories. Dividing by that denominator yields the per-trajectory effect. It is a hazard-estimation result; it is not, by itself, a causal estimate of the value of peer agreement.

Assigning the summed number to a differently named mechanism would change the experiment being described.

The gradient-boosted probe and isotonic calibration arms reduce utility relative to their matched logistic controls, by approximately 0.05694 and 0.06165 per trajectory. These negative effects constrain the claim that a more expressive probability model necessarily yields a better stopping policy. They do not prove that all nonlinear models overfit or that probability calibration is inherently harmful. Their effect depends on the fitted estimator, target, sample size, and policy threshold.

4.4 Token cap and numerical precision

The matched token-cap experiment compares 256 and 512 completion tokens for Mistral-Small-22B/GSM8K. Both arms have 454 losses among 1,500 trajectories under the recorded binary endpoint: the hazard policy has lower utility than never stopping. There are no discordant paired loss indicators. The empirical difference is zero, and the paired bootstrap is degenerate for that particular indicator.

A zero difference on this endpoint does not show that all answers, token counts, timings, or accuracy values are identical between caps. It also does not prove that truncation is absent from every model-domain cell. A scientifically valid negative result preserves the tested endpoint and avoids broadening it into a claim about every mechanism.

In the matched Qwen2.5-7B/GSM8K precision comparison, 418 of 1,500 step-two answers are correct under BF16 and 204 under 4-bit weights. The observed difference is 214/1,500, or 14.27 percentage points, with the recorded task interval approximately [11.13, 17.53] points. This is an absolute accuracy difference for that model, step, task

panel, and implementation. It is not a 14.3 percent relative reduction shared by every quantized model.

4.5 Ranking and stopping performance

Table 5. Stored causal detectors; common task-held-out development evaluation.

Configuration	Micro AUC	Task macro	Domain macro	Worst AUC	Step utility
Causal GRU	0.8743	0.8214	0.8102	0.6313	0.3264
Causal Fourier neural operator	0.8708	0.8172	0.8045	0.6152	0.3289
Selective SSM	0.8663	0.8116	0.7989	0.6078	0.3326
Causal residual TCN	0.8647	0.8088	0.7965	0.6057	0.3315
Causal RoPE transformer	0.8638	0.8085	0.7953	0.6079	0.3331
Truncated Beta mixture	0.8630	0.8140	0.7951	0.6204	0.3311
Five-expert causal MoE	0.8583	0.8005	0.7882	0.5955	0.3310
MoE plus hysteresis	0.8583	0.8005	0.7882	0.5955	0.3217
Task-grouped linear baseline	0.8294	0.7773	0.7612	0.6000	0.3244

The stored prefix-safe sequence comparison has a causal GRU micro AUC of 0.8743, task-macro AUC of 0.8214, and domain-macro AUC of 0.8102. Its worst domain is GPQA at approximately 0.6313. The contrast between pooled and worst-domain performance shows why a pooled score alone is inadequate for a deployment claim.

A causal RoPE transformer has lower pooled AUC than the causal GRU but slightly higher recorded micro step utility, approximately 0.3331 compared with 0.3264. Its token utility is approximately 0.3634 compared with 0.3631. The descriptive fold intervals overlap, so these stored summaries do not establish a statistically unique architecture

winner. They do show that ranking and stopping utility need not rank configurations identically.

Task-macro AUC is defined only for questions whose evaluated rows contain both correct and incorrect answers. There are 2,679 such tasks out of 2,948. Omitting the remaining tasks is appropriate for an undefined within-task ranking statistic, but the omission must be stated. Accuracy and utility can still include constant-label tasks.

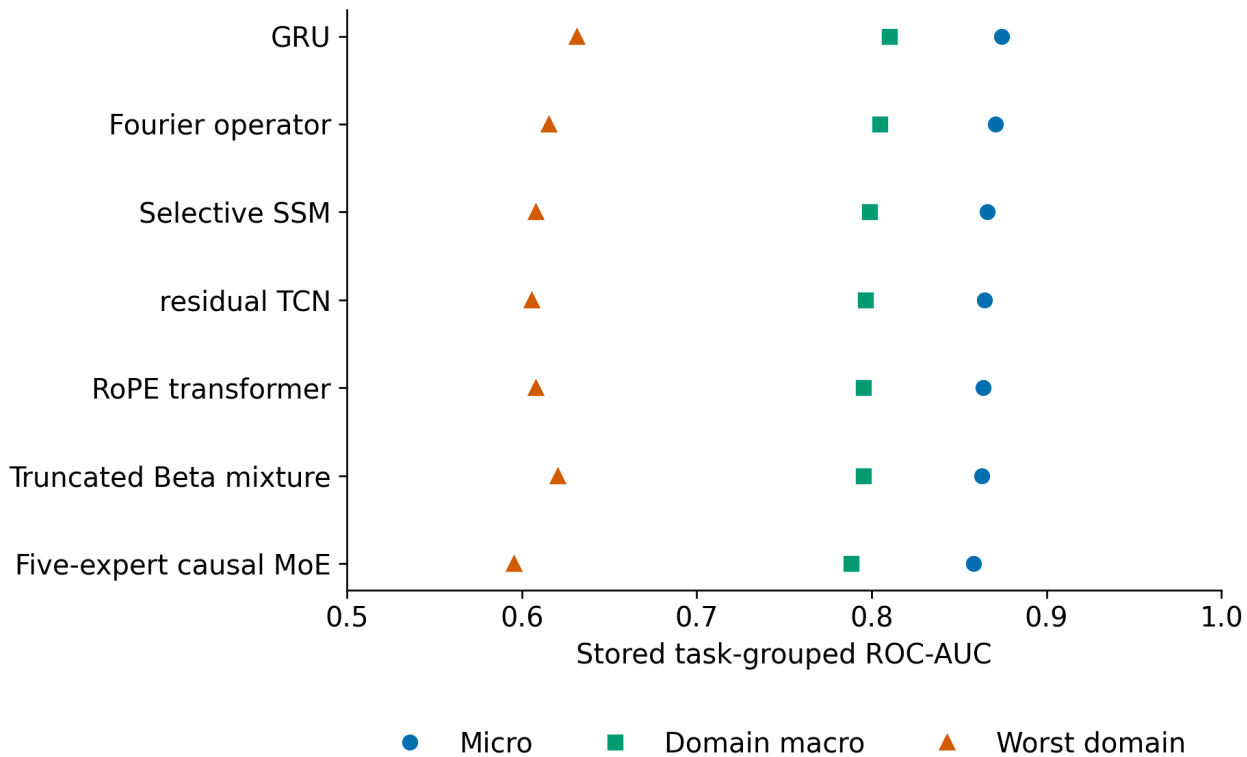


Figure 2. Micro, domain-macro, and worst-domain ranking differ substantially. Stored grouped outputs, not live-run results.

4.6 The stacked retrospective diagnostic

The historical stacked hybrid has stored AUC 0.955156, compared with 0.943223 for its reduced-feature LightGBM control. The observed lift is 0.011933. The stored task-bootstrap lift interval is [0.010384, 0.013480]. That interval describes the difference between two scores on this development evaluation; it is not an interval for the absolute stacked AUC.

Two design details limit its interpretation. First, a bidirectional sequence component reads all five saved steps and copies a trajectory score to earlier rows. A centered smoothing feature also uses later observations. The feature set therefore includes information unavailable to a live decision at an early step. Second, meta-training is not confined to the outer training partition of each reported scoring fold. Task-grouped scoring alone does not remove that upstream dependence.

The control also retains committee and independent-vote aggregates, so its label 'No Peers' does not denote a pure peer-free ablation. The numerical difference is a valid stored diagnostic of the implemented comparison, but it cannot isolate the causal value of peers or certify online correctness prediction. Bootstrap repetition cannot repair either information leakage or non-nested model selection.

The empirical proportion of positive lifts among 10,000 bootstrap draws is 100 percent. It is not a conventional p-value, a proof of superiority on all future questions, or evidence of independent generation seeds. Bootstrap randomness repeatedly resamples the same empirical development distribution. Reporting what it resamples is more informative than describing the resamples as 10,000 independent stress experiments.

4.7 Replay and failure interpretation

The historical Qwen2.5-7B/GSM8K replay counts 827,804 full-horizon completion tokens and 377,960 tokens up to the selected stopping steps. The implied saving is 54.34 percent. Accuracy declines from 70.53 percent to 64.20 percent, an observed loss of 6.33 points. The policy is fit and evaluated on the same 1,500 traces, so this result is a development diagnostic. Chapter 5 compares it with the newly implemented runtime experiments.

The newly recomputed paired failure audit covers all 75,965 sanitized variable-horizon trajectories. The archived hazard policy wins in 68,095 trajectories (89.64 percent), ties in 2,135 (2.81 percent), and loses in 5,735 (7.55 percent), relative to the full recorded horizon under the stored step utility. These are utility verdicts, not accuracy percentages. The audit reconstructs both binary endpoint labels from their utility and step cost, checks unique paired trajectories, and verifies that the taxonomy partitions every loss exactly once.

Table 6. Mutually exclusive archived-policy loss patterns; 5,735 utility losses among 75,965 trajectories.

Observed loss pattern	Count	Share of losses
No extracted candidate at the stop	196	3.42%
An earlier decision-eligible candidate was correct	460	8.02%
Step one was correct; no eligible pre-stop answer was correct	600	10.46%
First eligible correct answer arrived one step after the stop	1,706	29.75%
First eligible correct answer arrived at least two steps after the stop	2,773	48.35%

Every observed utility loss stopped on an incorrect candidate and ended with a correct full-horizon candidate. This pattern describes the current records; a correctness gain after an early stop can outweigh the accumulated cost. Among those losses, 196 stop states have an empty extracted candidate. Another 460 traces contain an earlier correct decision-eligible answer, and 600 contain a correct step-one answer but no correct eligible answer before stopping. The remaining losses first reach an eligible correct answer one step after stopping (1,706) or at least two steps later (2,773). The categories follow a fixed priority order, so an empty stop takes precedence over other properties of the same trace.

The larger late-repair group is 48.35 percent of losses. It does not mean that these traces invariably repaired at step five, or that all online predictors must fail. Horizons vary in this matrix. The recomputation uses frozen labels and does not execute a new regrade or train a new probe. Old probe AUCs embedded in the classification script's descriptive tags are excluded from this table. A failed linear predictor of a late repair is evidence about that predictor and its recorded features; it is not an information-theoretic impossibility proof for every possible online signal.

Passing thirty grader cases also does not establish that zero corpus errors are grading errors. The rigorous conclusion is narrower: the tested normalization and equivalence behaviors pass their regression suite, while the empirical tables use the frozen labels. An independent label audit remains a distinct source of evidence.

4.8 Empirical conclusions

The stored evidence supports cost-sensitive continuation decisions that vary by model and domain. Repairs can dominate early aggregate changes, later gains can fall below

their cost, and estimator changes can improve or worsen utility. It also supplies direct counterexamples to several overly broad claims: pooled AUC is not live policy accuracy, a zero token-cap contrast does not establish universal absence of truncation, and replay savings need not preserve accuracy.

The next scientific step is consequently specific. A policy must be fixed using permitted development information, evaluated without future-step features, executed in a loop that avoids further generation, and measured jointly on accuracy and all relevant costs. That is the subject of the online implementation and experiments in Chapter 5.

Chapter 5 Online stopping and computation trade-offs

5.1 Causal controller contract

The online controller accepts one newly completed response observation at a time. An observation contains its step number, candidate answer, parsing status, generated completion tokens, and any explicitly available confidence or instrumentation. The API has no field for the reference answer, the correctness label, a future response, or a score computed from the full trajectory. The controller stores its own prefix and rejects skipped, repeated, or out-of-order observations.

The policy enforces a minimum of two completed increments and a finite maximum horizon. Stopping closes the controller, so feeding a subsequent response is an error rather than an implicit restart. The decision records a selected answer and step, the reason for termination, the policy fingerprint, and measured latency. Invalid confidence, incomplete parsing, and future timestamps receive explicit handling instead of becoming undocumented numeric defaults.

Peer agreement is optional. When used, the API requires a complete, same-step roster whose barrier closes before the decision. An incomplete panel, a duplicate peer, or a vote that completes later cannot support a current stopping decision. These checks establish the declared event contract; timestamps alone cannot establish that an external caller supplied authentic observations. A fleet experiment must retain generation events and charge every peer's computation.

5.2 Policy choice and limitations

The first runtime policy is a frozen confidence-and-stability heuristic. Model-reported confidence is uncalibrated, and a stable answer can be wrong. Default selection carries forward the latest nonempty answer; confidence-drop and answer-wobble branches can retain a prior candidate. Neither retention branch fired in the reported live panels. The controller also provides fixed-step and never-stop controls under the same floor and horizon. These transparent policies make it possible to compare the engineering effect of termination separately from the value of correctness prediction.

The retrospective 0.955156 stacked score is excluded from the controller. Its bidirectional sequence score and future-step smoothing violate the runtime information contract, and a new causal implementation cannot inherit its AUC by reusing the name of the detector. Stored task-held-out predictions can support replay comparisons, but they are not a deployable fitted model unless the training artifact and its causal feature transformation are available and verified.

The heuristics are engineering choices rather than instances of the general Bellman optimum. A confidence drop may justify caution without proving that all future repair opportunities are exhausted. Oscillation can signal instability without distinguishing a wrong candidate from a correct one. Chapter 2's delayed-repair counterexample applies directly: immediate evidence against continuing does not, without structural assumptions, establish a globally optimal stop.

5.3 Generation integration

The generation adapter requests an increment, constructs its permitted observation, evaluates the controller, and requests the next increment only if the decision continues.

A stop therefore prevents future model calls. A caller cancellation is propagated through the adapter's generation lifecycle. Baseline and active policies use the same response-boundary and parsing conventions so the stopping comparison does not silently change the response contract.

This integration stops between complete reasoning responses. It does not claim that the risk heuristic identifies an optimal token inside a response. Cancellation and a common response-completion boundary can interrupt a call for engineering reasons, but they are different mechanisms from the policy's step-level decision. Completion-token savings refer to avoided response generation under this declared granularity.

The active and full-horizon arms are executed separately. For deterministic greedy generation, the evaluation checks whether their shared prefixes agree. Changing batch membership or kernels can affect numeric results, so deterministic sampling alone is not proof of identical prefixes. A measured mismatch must be reported, because it limits interpreting the full arm as an exact counterfactual continuation of the active arm.

5.4 Decision latency

Table 7. Controller-only latency on 100 saved math-question prefixes.

Scope	Decisions	Median ms	p99 ms	Maximum ms
All measured decisions	18880	0.00340	0.01300	0.43830
confidence_stability_v1	8880	0.00420	0.01534	0.43830
never	10000	0.00340	0.01060	0.11090

The latency experiment evaluates decisions on one hundred math-question prefixes, retaining per-step timings and the benchmark context. It reports the median, upper

quantiles, and maximum, with the number of measured decisions. A single fast average is not sufficient to establish a worst-observed under-ten-millisecond target. These measurements cover the controller call on already prepared observations, not the language-model forward pass or external peers.

Unit and adversarial state-machine checks complement this timing result. They verify the minimum floor, horizon termination, terminal closure, future-step independence, parsing guards, exact token accumulation, peer barriers, and cancellation. Their passing result establishes implementation behavior on the covered cases. It does not establish that the heuristic preserves accuracy on a distribution of language-model questions.

5.5 Live paired evaluation

Table 8. Actual paired local generation on 100 GSM8K questions.

Endpoint	Measured value
Questions	100
Full-horizon accuracy	6.00%
Stopped accuracy	6.00%
Accuracy difference	+0.00 percentage points
Full-horizon completion tokens	22,244
Stopped completion tokens	21,591
Full-horizon prompt tokens	142,498
Stopped prompt tokens	137,574
Completion-token saving	2.94%
Mean stopped step	4.85
Identical shared prefixes	94
Conservative paired 95% accuracy interval	[-4.29, +4.29] percentage points
Task-bootstrap saving interval	[0.85%, 5.63%]

The live experiment freezes the task panel, policy, and generation settings before opening evaluation outcomes. Public task records are supplied to generation separately from the reference-answer ledger. Grading occurs after the generation events are written. This separation makes the information boundary inspectable and prevents a runtime stopping decision from accidentally consuming the benchmark label.

The primary paired quantities are final-answer accuracy, selected-answer accuracy, generated completion tokens, executed response increments, prompt-token counts, and wall time. The report states whether auxiliary scoring or peer generations occur. The ratio-of-total completion-token saving is computed from actual generated counts in the two arms. A completion-token reduction is not automatically an equal reduction in wall time, energy, or total inference cost.

The sample is a local development-scale experiment on one available model. Even with a policy frozen before this run, tasks previously present in the project's corpus cannot be called an untouched external test set. A new run supplies genuine prospective execution evidence, but task novelty, model generalization, and independence from development selection remain separate properties.

The accuracy interval is reported alongside savings. An observed gain or zero difference is not a noninferiority proof unless the interval and a prespecified tolerance establish that claim. This thesis makes no unconditional assertion of zero accuracy loss. Likewise, a result from Qwen2.5-0.5B does not validate the earlier Qwen2.5-7B replay or the thirteen-model detector panel as a live system.

The completed heuristic run answers six of one hundred questions correctly in each arm. No paired correctness indicators change, but the conservative paired interval still allows an accuracy change between approximately -4.29 and +4.29 percentage points under the declared iid task-pair reference model. Completion tokens decline from 22,244 to 21,591, a saving of 2.94 percent; its descriptive task-bootstrap interval is approximately [0.85, 5.63] percent. Only six tasks stop by the confidence-and-stability rule, and ninety-four reach the maximum horizon. The mean stopping step is 4.85.

These observations fall well short of the roadmap's suggested thirty-to-forty-percent saving target.

The response contract is a substantial limitation. Only 88 of the 500 full-horizon responses are strictly valid complete JSON, and final baseline accuracy is six percent. The controller can retain a previously extracted nonempty answer when a later increment is empty, but that selection rule does not recover arbitrary malformed reasoning. All resulting failures remain included in the reported denominator. A study on a larger, protocol-capable model is needed before this experiment can support useful deployed answer quality.

Shared prefixes are exactly identical on 94 of the 100 paired tasks, rather than all of them. The remaining differences constrain a trajectory-specific counterfactual interpretation even though both arms use greedy sampling and the same frozen generation adapter. Measured completion tokens are supplemented by prompt tokens (142,498 versus 137,574), model time (1,127.05 versus 1,099.21 seconds), padded prefill slots (231,856 versus 222,781), and padded decoding slots (57,856 versus 56,128). Prompt plus completion tokens fall by 3.39 percent. There are no auxiliary verifier or peer generations. These quantities describe different costs and are not interchangeable efficiency percentages.

5.6 Adversarial questions and traps

Table 9. Actual paired local generation on the 20-question trap bank.

Endpoint	Measured value
Questions	20
Full-horizon accuracy	5.00%
Stopped accuracy	5.00%
Accuracy difference	+0.00 percentage points
Full-horizon completion tokens	4,736
Stopped completion tokens	4,578
Full-horizon prompt tokens	27,641
Stopped prompt tokens	26,143
Completion-token saving	3.34%
Mean stopped step	4.75
Identical shared prefixes	20
Conservative paired 95% accuracy interval	[-19.68, +19.68] percentage points
Task-bootstrap saving interval	[0.00%, 8.80%]

The adversarial bank contains twenty independently specified mathematical questions, with separate public prompts and evaluation labels. It includes changing percentage bases, harmonic average speed, conditional probability, dependent sampling, inclusive endpoint counts, reciprocal rates, exponent precedence, answer-contract distinctions, and irrelevant details. Exact short derivations accompany the evaluation labels so they can be checked independently of model output.

This bank is designed to probe common reasoning traps, not to estimate the worst-case failure rate of every possible attack. Some classic puzzle forms may have appeared in model pretraining even though these exact project records are newly written. The measured outcome is therefore robustness on the frozen bank under this model and prompt protocol. The software's malformed-input and forged-peer tests remain a separate engineering evaluation.

The completed heuristic trap run answers one of twenty questions correctly in each arm. Completion tokens decrease from 4,736 to 4,578, a saving of 3.34 percent; all twenty shared prefixes match exactly. Only two tasks stop by stable confidence, and the other eighteen reach the horizon. The mean stopping step is 4.75. Full-horizon strict JSON validity is fourteen of one hundred responses. The conservative iid-reference accuracy difference interval is approximately $[-19.68, +19.68]$ percentage points, and the descriptive token-saving interval is $[0.00, 8.80]$ percent. This small, low-accuracy panel provides no strong accuracy-preservation or broad adversarial-robustness claim.

A wrong but stable high-confidence answer is particularly informative: it demonstrates a limitation of the confidence-and-stability heuristic even when the controller functions exactly as intended. A late repair after termination similarly tests the stopping approximation. The evaluation retains these adverse outcomes rather than choosing a new threshold on the same questions and relabeling the adjusted result confirmatory.

The conservative paired accuracy interval uses simultaneous 97.5-percent Clopper-Pearson intervals on the improvement and worsening probabilities. A union bound gives at least 95-percent simultaneous coverage under iid task-pair sampling, without assuming the two discordance categories are independent. Subtracting the interval endpoints then bounds their difference. The task-bootstrap saving interval resamples

paired questions and recomputes the ratio of total tokens. Because this trap bank is handpicked, its interval is a conditional reference calculation and resampling diagnostic; it does not establish randomized adversarial-population coverage.

5.7 Deployable prefix probability model

The second runtime policy uses a pair of fitted, portable probability heads. One estimates correctness of the latest nonempty candidate available in the observed prefix, $\hat{q}_t$. The other estimates correctness of the candidate selected by the same rule after one additional increment, $\hat{p}_{t+1}$. Direct prediction of the latter avoids dividing the small sample into separate repair and corruption strata. The controller recomputes the estimated one-step gain as

$$\hat{\mu}_t = (v + p)(\hat{p}_{t+1} - \hat{q}_t) - \lambda.$$

The frozen policy uses $v = 1$, $p = 0$, and $\lambda = 0.05$. It stops at the first nonpositive estimated gain after the two-step floor, otherwise at step five. It does not query a next-step predictor at the terminal horizon. This is a myopic fitted rule. No persistence assumption or learned Bellman continuation law is established for these generations.

Training uses all 1,500 archived Qwen2.5-0.5B GSM8K-training and MATH trajectories, containing 7,500 rows. A fixed hash of public task identity allocates 902 tasks to fitting, 322 to calibration, and 276 to evaluation. Standardization and unweighted logistic coefficients are fit only on training tasks. Each one-dimensional Platt logistic calibrator uses only calibration tasks. The evaluation tasks enter neither fit. All rows from a task share its partition, and the live hundred-task and twenty-trap public identities are explicitly excluded. Features use only current and prior observations, including step, token count, answer changes, thought-text summaries, and public domain.

The training targets match runtime answer selection. Reconstructing candidates with the declared parser produces 4,269 candidate-string disagreements with the saved candidates, so the reconstructed candidates are regraded against the archived references using the existing verification rule. Only 43 row-level correctness labels change. This new training-label version and a row-level audit are separate artifacts; the frozen source corpus and the primary empirical chapter's stored labels remain unchanged. No trajectory is filtered because of later candidate agreement. Selecting whole trajectories by agreement of all five responses would condition an early-step evaluation on future observations and change the eligible population.

Table 10. Deployable probability heads on 276 archived held-out tasks; repeated rows are not independent trials.

Target	Held-out rows	AUC	Brier	ECE (10 bins)
p_next	1104	0.6923	0.0997	0.0209
q_current	1380	0.7101	0.0976	0.0222

The fitted artifact serializes both scalars, coefficient vectors and calibrators, allowing standard-library probability evaluation without scikit-learn at runtime. Its byte hash is checked by the controller against the frozen policy. The calibrated held-out Brier scores improve slightly over their uncalibrated versions (0.09854 for the current target and 0.10068 for the next target). Those aggregate scores and ten-bin calibration summaries are marginal development diagnostics, rather than certified conditional probabilities for a live prefix.

Transport from the archive remains a material limitation. None of the 7,500 archived outputs satisfies the live strict JSON contract. Consequently, confidence and strict-

parsing features have no observed archive variation, while the live JSON prompt creates a different feature distribution. Archived token counts also omit emitted EOS tokens that the live ledger includes. The evaluation must disclose these differences rather than describe task-disjoint archive calibration as live calibration.

On the 276 held-out archived tasks, the learned rule stops 275 times at step two and once at step three. Its accuracy is 11.96 percent, compared with 10.51 percent for the full horizon; the task-bootstrap paired accuracy interval includes zero, approximately $[-1.45, +4.35]$ percentage points. Replayed completion-token saving is 51.29 percent, with an approximate interval $[49.31, 53.09]$ percent. The near agreement with a fixed-two-step policy means this evaluation demonstrates little additional stopping value from the learned heads. It does establish a reproducible, causally computed fitted artifact that can be evaluated prospectively.

5.8 Actual trained-policy evaluation

Table 11. Actual trained-policy generation paired with the existing 100-question live baseline.

Endpoint	Measured value
Questions	100
Full-horizon accuracy	6.00%
Stopped accuracy	7.00%
Accuracy difference	+1.00 percentage points
Full-horizon completion tokens	22,244
Stopped completion tokens	9,674
Full-horizon prompt tokens	142,498
Stopped prompt tokens	43,874
Completion-token saving	56.51%
Mean stopped step	2.00
Identical shared prefixes	100
Conservative paired 95% accuracy interval	[-4.27, +6.21] percentage points
Task-bootstrap saving interval	[54.83%, 58.01%]

Table 12. Actual trained-policy generation paired with the existing trap-bank live baseline.

Endpoint	Measured value
Questions	20
Full-horizon accuracy	5.00%
Stopped accuracy	5.00%
Accuracy difference	+0.00 percentage points
Full-horizon completion tokens	4,736
Stopped completion tokens	2,268
Full-horizon prompt tokens	27,641
Stopped prompt tokens	8,115
Completion-token saving	52.11%
Mean stopped step	2.00
Identical shared prefixes	20
Conservative paired 95% accuracy interval	[-19.68, +19.68] percentage points
Task-bootstrap saving interval	[45.86%, 57.17%]

The trained policy generates its own actual stopped trajectories. It is paired with the already executed full-horizon baseline, whose manifest, model bytes, public prompts, generation adapter, batch order and results are linked by hash. Reusing an actually generated baseline avoids unnecessary repeated GPU work and does not turn the newly executed learned arm into saved-prefix replay. The measured shared-prefix agreement and physical accounting must nevertheless be reported, because earlier baseline execution and later active batching can produce numeric differences.

The predictor artifact and policy are frozen before learned generation. No live or trap outcomes enter its coefficient, calibrator, feature, cost or threshold fitting. The comparison remains a development experiment on one small model and two declared panels. It does not deploy the retrospective stacked hybrid, establish per-instance safety, or prove that the learned rule improves on a cheap fixed-step policy.

The completed trained-policy run stops at step two on all one hundred GSM8K questions. It answers seven correctly, compared with six at the full horizon, and saves 56.51 percent of completion tokens (22,244 versus 9,674). The descriptive saving interval is approximately [54.83, 58.01] percent. The conservative paired accuracy-change interval is approximately [-4.27, +6.21] percentage points, so the observed one-point improvement does not demonstrate superiority or accuracy noninferiority. Every shared prefix matches exactly. Since every task stops at the same step, this realized policy is equivalent to a fixed-two-step budget on the collected panel; the experiment supplies no evidence of useful adaptive stopping beyond that budget.

The trained trap run also stops at step two on all twenty tasks, with one correct answer in both arms. Completion tokens decline from 4,736 to 2,268, saving 52.11 percent, with a descriptive task-resampling interval approximately [45.86, 57.17] percent. All twenty shared prefixes match. The handpicked-bank accuracy interval retains the broad iid-reference bounds of approximately [-19.68, +19.68] percentage points. The large compute reduction accompanies weak absolute answer accuracy and does not establish general adversarial robustness.

Accounting includes repeated prompt processing: the trained main arm uses 43,874 prompt tokens instead of 142,498, and the trap arm uses 8,115 instead of 27,641. Prompt-plus-completion savings are approximately 67.50 and 67.93 percent. Measured

main model time is 343.25 seconds rather than 1,127.05, and trap model time is 74.12 rather than 245.14. Padded prefill and decoding slots are separately recorded, with no verifier or peer generations. These are observations on this execution environment; they do not guarantee the same latency reduction under another model or batching regime.

On one hundred actual baseline prefixes repeated twenty times, the trained controller executes 4,000 decisions with median 0.059 ms, p99 0.3143 ms and maximum 0.9369 ms. This benchmark includes feature extraction, both probability heads, calibration, validation and drift computation. It excludes language-model loading, generation, tokenization and peer waits. The measured decision overhead satisfies the under-ten-millisecond target at the observed maximum.

5.9 Replay and Pareto analysis

Table 13. Frozen variants replayed on 1,500 saved trajectories. These are development comparisons.

Policy	Accuracy	Completion tokens	Saving	Nondominated
never	9.53%	418,845	0.00%	False
fixed_2	9.60%	204,011	51.29%	True
fixed_3	9.93%	278,444	33.52%	True
fixed_4	9.73%	349,254	16.61%	False
confidence_80	9.60%	374,129	10.68%	False
confidence_90	9.60%	377,172	9.95%	False
confidence_95	9.53%	382,981	8.56%	False

A development replay compares the frozen heuristic with fixed-step policies and the full horizon on already saved trajectories. It is useful because all policies can be evaluated against the same recorded continuations at low additional cost. Its counterfactual validity still requires that stopping does not change the preceding generation protocol. It cannot measure live controller overhead, asynchronous batching effects, or energy.

The accuracy-cost plane reports each policy's observed accuracy against its measured or replay-counted completion tokens. A policy is dominated when another policy achieves at least as much accuracy with no greater cost and improves at least one quantity. Nondominated policies form the empirical Pareto set. A point on that set is not automatically preferred: choosing among points requires a declared utility weight or an accuracy constraint.

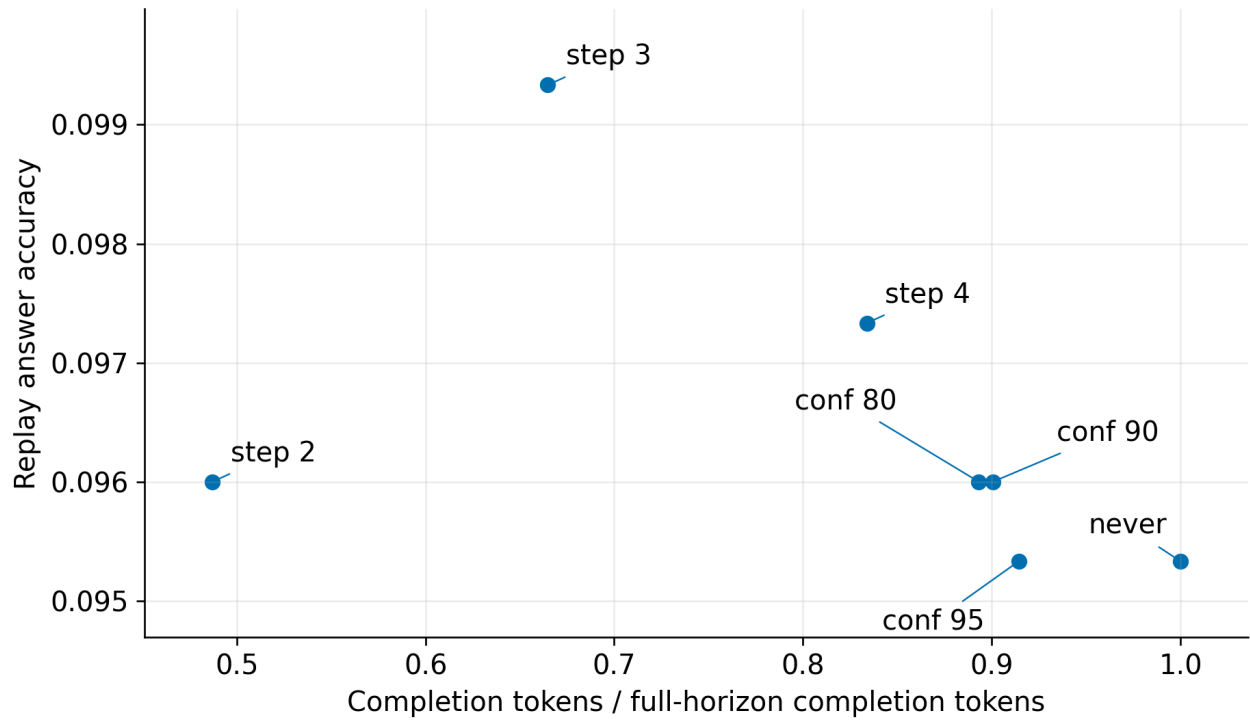


Figure 3. Development replay accuracy against completion-token cost. Labels identify fixed-step and confidence variants; observed nondominance is sample-specific.

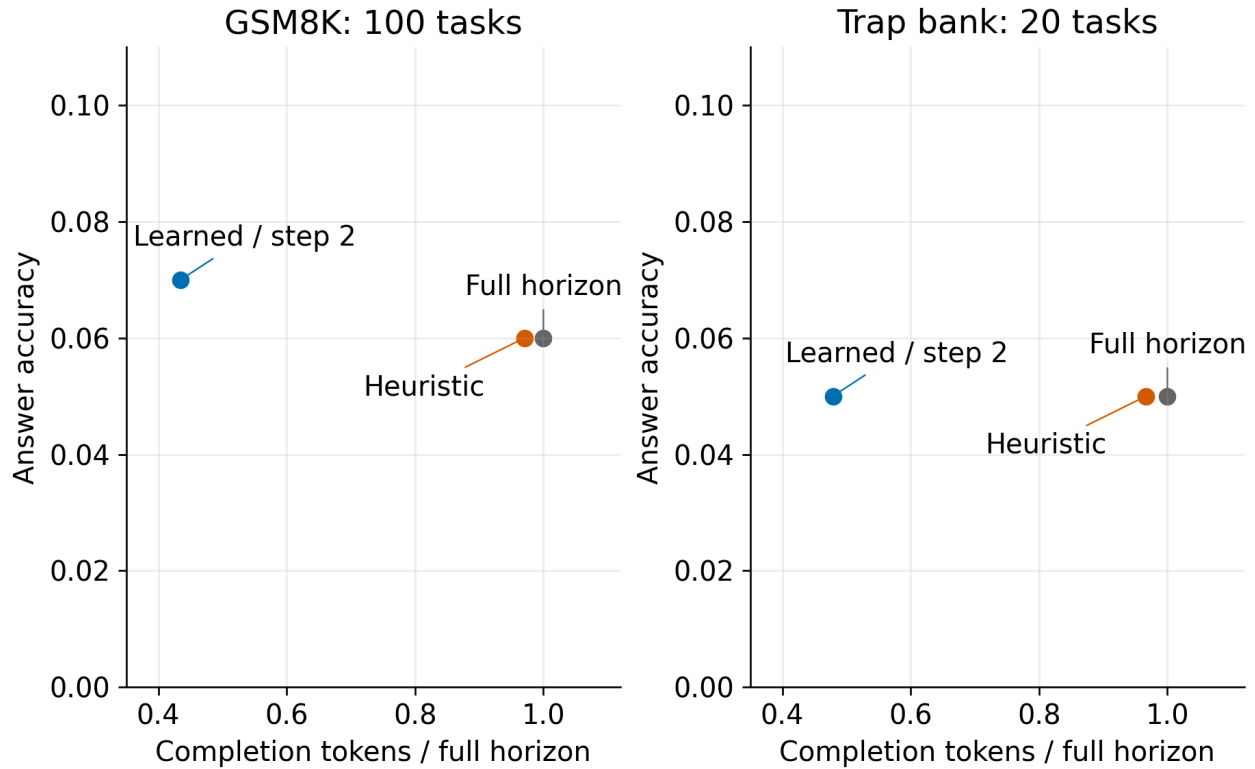


Figure 4. Actual generated arms on two development panels. Each learned arm reuses its panel's actually generated full-horizon baseline and stops at step two on every task. The plotted point therefore supplies no evidence of adaptation beyond a fixed-two-step budget. Point estimates omit uncertainty, which is reported in Tables 8, 9, 11 and 12.

Policy variants chosen after examining the same replay outcomes remain development comparisons. The diagram does not certify which point will be nondominated on new tasks. Confidence intervals and a frozen selection rule are necessary for the next external evaluation. A larger computation budget can also move a point by producing different answers, so a comparison should specify whether it changes the maximum response horizon or only the stopping rule.

5.10 Online conclusions

The online work establishes a causal runtime interface, direct control over future response scheduling, and reproducible measurements of decision cost and answer-cost trade-offs. The live and replay results have distinct roles. Live events demonstrate actual avoided generation; replay provides broader development comparisons on a shared stored panel; constructed tests verify the event contract.

The evidence supports reporting those results at their actual scope. A fast stopping decision and a positive completion-token saving are useful engineering outcomes. Preserved accuracy, optimality, safe calibration, and transfer to another model are further claims that require their own evidence. Keeping those requirements separate makes the controller and its limitations suitable for critical mathematical and scientific review.

Chapter 6 Discussion and limitations

6.1 Scientific interpretation

The central finding is that another reasoning increment has an uncertain and cost-dependent value. A question can benefit from revision, suffer a later answer corruption, or remain unchanged while consuming computation. The exact binary-transition decomposition makes these cases measurable. The optimal-stopping formulation explains why their relevance depends on future opportunities as well as the next step.

This combination supports a stopping problem, but not a universal stopping step. A panel curve mixes question difficulty, model configurations, and sampling outcomes. Even a stable population crossing need not identify the best decision for each prefix. A deployed system needs a conditional predictor, a continuation model, or an explicitly acknowledged heuristic. Its performance must be measured under the information available at runtime.

The distinction is also practical. A simple fixed-step policy can satisfy causal execution and save tokens without claiming to recognize correctness. A more elaborate detector may rank answers well while consuming peer or scoring computation that offsets some savings. Complexity is justified by its measured contribution to the final objective, not by the sophistication of its architecture.

6.2 Correctness labels and mathematical validity

The observable correctness state is defined by a reference grader. It is not a direct observation of mathematical understanding. Numeric extraction can fail on wording, symbolic equivalence can fail on domain assumptions, and an MCQ option can be

parsed correctly while its rationale is wrong. These issues can affect both training targets and estimates of repair and corruption.

The regression tests improve confidence in particular parser behaviors, but a comprehensive label-validity claim would require stratified human adjudication or an independent trusted verifier. Difficult symbolic cases deserve disproportionate attention because surface normalization may identify expressions that differ on an excluded domain or fail to identify mathematically equivalent expressions. The audit should distinguish parsing failure, benchmark-reference issues, and genuine reasoning errors.

Stored labels are preserved in this draft so the estimand stays fixed. A new grading version should create new label columns or new artifacts, publish the row-level changes, and recompute all dependent tables. Reporting both versions where their differences matter makes the scientific effect of the correction reviewable. It is preferable to silently replacing labels and retaining numerical summaries calculated from their predecessors.

6.3 Theory and deployable beliefs

The correctness posterior, repair probability, and corruption probability are conditional expectations under the true data-generating distribution. A fitted classifier is an estimate of one of those quantities only under additional assumptions about its inputs, training target, sampling distribution, and calibration. A model trained with class-balanced loss can be useful for ranking while targeting a reweighted distribution rather than the natural deployment posterior.

The theorem in Chapter 2 does not certify those estimates. It states what an optimal policy would do if the relevant conditional expectations were known. The finite-system

checks verify algebra and dynamic-programming behavior on specified examples. They do not turn an estimated model on a development corpus into the true conditional continuation law of an unseen language-model deployment.

Similarly, a confidence interval over questions is not a confidence sequence along a single reasoning trajectory. The former describes a population estimator under a resampling design. The latter needs a valid sequential concentration or martingale construction for observations generated by the actual process. Calling a threshold 'safe' does not supply that construction. This thesis avoids a per-instance error guarantee that the recorded evidence cannot support.

6.4 Generalization and adaptation

The project contains a broad but fixed panel of models and domains. A single generation seed per setting leaves seed variability largely unmeasured. Benchmark questions may have appeared during language-model pretraining, and task-held-out detector folds cannot eliminate that possibility. The collection's revision prompts, temperatures, response limits, and grading conventions further define the population to which its results apply.

Repeated algorithm development on the same corpus introduces selection effects. Even an outer task holdout can become a development set when many configurations are repeatedly compared and the best result determines the next experiment. A final confirmatory study needs a frozen policy, a locked feature contract, and an untouched evaluation panel. Its report should record any deviations from the prespecified experiment.

Domain stress tests already suggest that transfer is weaker than pooled internal evaluation. GPQA is a limiting domain in the stored causal sequence results. A new benchmark, another model family, or an adversarial prompt distribution can shift the meaning of confidence and agreement. Deployment requires monitoring and reevaluation rather than assuming the four-domain panel establishes indefinite generalization.

6.5 Costs beyond completion tokens

Completion-token reduction is a useful direct measure when a controller prevents responses from being generated. It is still only part of total inference cost. Repeated prompt processing, verifier forward passes, peer-model generations, batching effects, synchronization, network delay, and controller execution all contribute. Wall time and energy need direct instrumentation if they are the claimed outcomes.

Peer agreement particularly changes the comparison. If the baseline generates one model and the stopping system launches thirteen models, a shorter target trajectory may nevertheless cost more in aggregate. Peer signals are valid runtime information only after the peers have produced them, and their generation cost must be charged. A synchronized panel and an asynchronous deployment are different policies with different latency and cost.

The controller latency benchmarks measure decisions on already available observations. The learned benchmark includes prefix feature extraction, both probability heads, calibration and drift calculation. They exclude model loading, response generation, tokenization, peer waits and network communication. An under-ten-

millisecond decision satisfies the measured engineering target, but cannot by itself imply that total response time is unchanged.

6.6 Adversarial behavior and answer selection

A stable answer can be stably wrong. Several models can agree because they share a misleading interpretation. A correct first answer can be replaced at the mandatory second step. A trajectory can look unpromising for several increments and then repair. These cases are useful adversarial tests because they challenge different assumptions behind a stopping heuristic.

The software tests use constructed prefixes to check the decision contract and failure behavior. A live adversarial evaluation additionally needs language-model generations on prespecified trap questions, with independent answer keys and costs. Passing a synthetic state-machine test should not be described as passing that empirical challenge. Both forms of evidence are retained with their actual interpretation.

Answer selection is an extension worth evaluating separately. A controller could retain several previous candidates and select one at termination using permitted evidence. That action can recover some opportunities lost by a mandatory step floor. Selecting the retrospectively correct candidate using the reference answer, however, is an oracle comparator rather than a deployable policy. The final study should report stopping and answer selection as distinct interventions.

6.7 Evidence needed for stronger claims

To establish a claim of savings without meaningful accuracy loss, the experiment needs a prespecified noninferiority margin, a justified sample size, and a paired evaluation of a

frozen policy against the full-horizon baseline. A confidence interval must exclude losses larger than that margin; an observed zero difference is insufficient. Independent evaluation tasks and generation seeds would strengthen transfer and reproducibility evidence.

To establish a genuinely optimal learned boundary, the state representation and estimated transition law need support. A one-dimensional correctness estimate is not automatically sufficient for all future revision dynamics. Demonstrating predictive sufficiency would require showing that omitted prefix information does not materially improve the relevant continuation prediction under the target distribution. In practice, the thesis's general full-history formulation is more defensible than assuming sufficiency without testing it.

To establish numerical reproducibility of a historical tournament, the original training environment, feature construction, split membership, fitted artifacts, and stochastic controls need to be recovered or rerun. The data freeze and current environment lock are valuable progress, but they cannot reconstruct unavailable historical facts. A re-analysis with a fully captured new environment would be a new version of the experiment, not proof that the original version had every required record.

6.8 Thesis completion and academic review

The manuscript provides a complete argument across introduction, theory, methodology, empirical evidence, online engineering, and limitations. Its status is a research draft. Committee feedback can change definitions, emphasis, analyses, and presentation. The revision record should preserve each requested change, the

associated evidence, and its resolution; draft versions should be based on actual revisions rather than advancing labels without new review.

An oral defense, committee signatures, electronic-thesis acceptance, and degree clearance are academic events. They cannot be established by a successful code run or an assistant's statement. The roadmap retains those milestones as pending until the student and institution supply the corresponding evidence. A committee-delivery package can be prepared now, but its transmission is a separate action and does not substitute for review.

The final archival PDF also has separate requirements. The current draft follows the documented margins, spacing, typography, title-page information, and numbering conventions as far as its build and visual checks establish them. PDF/A conformance must be checked by a dedicated validator after final revision. Embedded fonts or a filename ending in PDF do not prove conformance. The university's final submission procedure should be completed only after the defense and final edits [12].

6.9 Conclusion

Cost-aware stopping offers a precise way to study useful and harmful continuation in reasoning models. The project's experiments show model- and domain-dependent trade-offs, demonstrate the importance of matched comparisons and causal feature availability, and expose the limits of retrospective detector scores. The mathematical formulation supplies a valid general optimum and a clear explanation of when simpler drift rules can fail.

The resulting controller makes an implementable prefix-only decision and provides a basis for measured live experiments. The thesis's strongest conclusion is conditional

and operational: computation should be allocated according to the expected value of permitted continuation, and the evidence for that allocation must jointly report answer quality, incurred cost, and the information available at the decision. Stronger universal or safety claims remain subjects for independent experiments rather than assumptions embedded in the current result.

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Appendix A Reproducibility and evidence sources

Run the following commands from the repository root. The environment provenance distinguishes the current workstation from the historical training environment. A successful verification confirms the frozen evidence identity under the chosen byte rule; it does not regenerate the language-model corpus.

```
python tools/recompute_thesis_evidence.py
python research/tests/test_graders.py
python research/tests/test_boundary_floor.py
python research/tests/test_mathematical_foundations.py
python research/tests/test_online_controller.py
python -m pytest research/tests/test_prefix_stopping_model.py
research/tests/test_learned_online_controller.py
research/tests/test_live_uncertainty.py -q
python tools/summarize_failure_audit.py
python tools/analyze_live_stopping_uncertainty.py
research/outputs/semester2/online_stopping_20261002
research/outputs/semester2/online_stopping_20261002/adversarial_live
research/outputs/semester2/online_stopping_20261002/learned_main
research/outputs/semester2/online_stopping_20261002/learned_adversarial
python tools/build_master_thesis.py
```

Data-freeze and live-evaluation commands are documented in the corresponding implementation reports. Use their exact model-local path and frozen task/policy arguments when reproducing a generation experiment. The thesis build requires the pinned KaTeX dependency recorded in `tools/thesis_pdf_package.json`, the Python packages used by the builder, and a local Chrome or Edge executable. It does not fetch external mathematical assets while rendering.

Table 14. Claim evidence sources.

Claim or output	Authoritative repository source
Tournament corpus	data_manifest_v1.json and ultimate_tournament_manifest.json
Boundary and matched effects	research/outputs/thesis_v1/evidence/*.csv
Proofs and exact finite checks	research/mathematical_foundations.md; test_mathematical_foundations.py
Controller execution	online_stopping_controller.py; online_generation.py
Latency and replay	research/outputs/semester2/online_stopping_20261002/
Adversarial bank	research/adversarial_tasks_v1.jsonl; adversarial_gold_v1.jsonl
Archived policy failures	research/reports/thesis_failure_audit_v1/audit_summary.json
Deployable prefix predictor	research/outputs/semester2/prefix_model_v1/
PDF authoring	tools/build_master_thesis.py; tools/render_thesis_math.mjs

The standardized tournament manifest lists the mandatory fifty-two trace files. Supporting evidence adds the matrix, paired arms, cached estimator comparisons, stored OOF predictions, analysis implementations, and newly collected events. This source selection is explicit, so a reader can distinguish a fingerprint of selected evidence from a census of every incidental file in the repository.

Appendix B Claim scope and mathematical review

Claim	Evidence required	Current interpretation
Binary repair-corruption identity	Common transition panel or conditional probability proof	Exact decomposition in Chapter 2
General optimal stopping	Full conditional continuation law and all costs	Finite-horizon theorem; not a calibrated learned deployment
First drift crossing is optimal	Pathwise persistence after the crossing	Conditional theorem; not established universally by the empirical curves
0.955156 AUC	Stored historical tournament result	Retrospective non-nested diagnostic
Causal detector ranking	Task-grouped causal sequence outputs	Internal development evaluation
Replay completion-token saving	Frozen traces and selected prefixes	Counterfactual development quantity
Live completion-token saving	Actual generation events from both arms	Model-, task-, and protocol-specific measurement
No meaningful accuracy loss	Prespecified tolerance and adequate paired confidence interval	Not inferred merely from observed equality
Grader has no errors	Independent corpus adjudication	Thirty regression cases do not prove this
Thesis accepted and degree cleared	Committee and institutional records	Pending human and institutional milestones

The proof audit explicitly checks the filtration, hidden correctness target, measurability of stopping events, finite-horizon integrability, cost accounting, and the difference between conditional beliefs and sample averages. Exact finite-system tests supplement those proofs by checking enumerated policy values and counterexamples. Neither a test suite nor a plot substitutes for a general mathematical argument.

Appendix C Revision and final submission

The roadmap's two committee cycles require actual feedback. For each cycle, record the reviewer comment, affected statement, evidence or proof needed, resolution, and verification. Do not identify a draft as approved without the corresponding approval. The planned defense deck, paper, mock defenses, oral defense, signatures, library deposit, and registrar clearance remain traceable in the milestone audit.

The electronic-thesis draft uses one-inch digital margins, a body font of at least ten points, double-spaced body text and abstract, Roman front-matter numbering, and Arabic main-text numbering. These are formatting targets derived from the current Sheridan Libraries requirements [11]. Final PDF/A export and conformance validation follow the final committee-approved revision. No conformance or submission claim is made for an ordinary review PDF.

Appendix D Paired accuracy uncertainty

For a task sampled under a common independent and identically distributed task law, let I indicate an incorrect baseline answer repaired by the active policy and let W indicate a correct baseline answer corrupted by it. The population accuracy difference is $\delta = \pi_I - \pi_W$, where $\pi_I = \Pr(I = 1)$ and $\pi_W = \Pr(W = 1)$. The two indicators are mutually exclusive within a task; the analysis does not assume their independence.

Across n independent task pairs, each marginal discordance count is binomial. Construct an exact two-sided 97.5% Clopper-Pearson interval $[L_I, U_I]$ for π_I and a separate interval $[L_W, U_W]$ for π_W [17]. Each interval has noncoverage probability at most 0.025. The union bound therefore gives simultaneous coverage at least 0.95,

irrespective of the dependence between the two counts. On that event, subtraction yields

$$\delta \in [L_I - U_W, U_I - L_W].$$

This is the conservative 95% paired interval reported for the live panels. Exact finite multinomial outcome enumeration supplements the coverage argument; it does not replace the binomial assumptions. With no observed discordances, both marginal lower bounds are zero and both upper bounds equal $1 - 0.0125^{1/n}$. Thus observed equality gives a nonzero uncertainty interval, rather than evidence of exact equivalence. No accuracy tolerance or noninferiority hypothesis was prespecified.

The twenty hand-selected adversarial tasks are a fixed challenge bank. An interval calculated using an independent-task reference model does not provide randomized coverage for an adversarial population. Completion-token intervals instead use a paired task bootstrap of the ratio of total token differences to total baseline tokens. These are descriptive resampling intervals, rather than exact finite-sample certificates.